

Four workbenches and where they meet

Erdős–Straus, Collatz, Erdős Problem 817 and Yang–Mills,
with the Navier–Stokes, S^6 and Jacobian-map material:
a reader of their results, with proofs

An audit and reading prepared by Claude (Anthropic),
model `claude-opus-5-5` (Opus 5.5), at maximum reasoning effort

Workbenches: KokunoYumeto and the contributors credited in each section

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Abstract

This reader sets out what four public workbenches have established, as found in an audit carried out on 25–26 September 2026, and where they meet each other and the split-zero programme around the Riemann zeta function. The workbenches concern the Erdős–Straus equation $4/p = 1/x + 1/y + 1/z$; the Collatz map; Erdős Problem 817 on subset sums without arithmetic progressions; and the Hamiltonian lattice Yang–Mills theory, together with the material the Yang–Mills workbench holds on the Navier–Stokes equations, on a manuscript that constructs a compact complex threefold which it states is diffeomorphic to S^6 (not checked here), and on a counterexample to the Jacobian conjecture. Sections 1–4 state each workbench’s results; the central ones are proved in full here, several of them in greater generality than the workbench states, and the others are listed with their status or by a pointer to the first-pass reports. Each section ends with the negative results that were checked, each within the exact scope of what fails. Section 5 states every intersection found between the workbenches, and with the zeta programme, as a typed map with its status, and records three patterns that recur. Nothing in this reader proves or disproves the Erdős–Straus conjecture or the Collatz conjecture, or establishes or refutes a mass gap for the continuum Yang–Mills theory, and none of the workbenches claims to.

About this reader

What was read. The audit read four public repositories maintained by KokunoYumeto, at fixed commits: `erdos-straus-foundation` at `d20b32e` (23 September 2026); `collatz-workbench` at `caf04ff` (16 September); `erdos-problem-817-workbench` at `dbd0a93` (16 September); and `yang-mills-interacting-workbench` at `fa79faf` (21 September), which also holds the Navier–Stokes (NS), S^6 and Jacobian-attribution material. Each workbench is larger than what was audited; Section 6 lists what was not checked, and each result below says how far it was read.

How the audit was done. For each workbench a first-pass audit was made by a Claude subagent: it restated each result with its hypotheses, followed the written proof step by step and wrote independent checks; the Collatz and Yang–Mills first passes also re-ran the workbenches’ own checkers. The author of this reader then re-ran the first-pass Python scripts except one 168-second script (identical outputs apart from timing lines; the first-pass C searches were not re-run, and a separate search of the author’s replaces them for small n), wrote a further independent check script from the definitions alone, re-derived the proofs given here, looked for statements that hold more generally than stated, and wrote five audit notes (43_–47_). Claude instances that had written none of this refereed the notes, then this reader, then this reader twice more for understatement and missed generality as well as for overstatement. None found an error in a written-out proof; the last pass found one false statement of generality, three results of a workbench that had been credited to the audit, and a published cycle bound stronger than the one derived

here; all are corrected. Every finding (Appendix A) was checked against the sources before it was applied. The notes, the first-pass reports, the scripts and their outputs are on the branch `claude/claude-ab-grind-20260925` of the zeta repository.

Who did what. The results are the workbenches' and the contributors' they credit; each section opens with the workbench's own credit line. The audit's own contributions are the re-derivations, the checks, and the generalisations marked Generalised here. Among these are the range $x < p/2$ of the Erdős–Straus shell criterion, extended from the workbench's primes $p \equiv 1 \pmod{12}$ to every odd prime; the order-2 gap theorem with the threshold lowered to $g^2 \geq 4.0187$ using audited inputs only (the workbench itself has the order-1 case and order-2 thresholds 4.1156 and, with a further computed input, 3.8260); the girth form of the Casimir inequality; and the lifting of the $k = 4$ construction. They also include the proofs of the two-shell survivor classes and of the reduction modulo 840; the Eliahou-type cycle bounds from a verification to 2^{33} made here and from Barina's 2021 range, which are weaker than the published bound of Hercher and Barina, also stated (Section 2); a Pocklington certificate for an Erdős–Straus prime; the exact small values of g_4 (Section 3); and the statement of the intersections (Section 5). No novelty is claimed for any of them.

Status labels.

- Proved here: the proof is given in full in this reader and was checked.
- Generalised here: a workbench result stated and proved here under weaker hypotheses than the workbench states.
- Audited: the proof is in the workbench; it was read in full, every step was checked, and independent checks were run; it is not reproduced here.
- Audited (certificate): the finite certificate on which the proof rests was recomputed independently, and the deduction from it was read.
- Audited in part: the parts named were checked; the rest is located.
- Conditional: audited or proved here, given an input that was not re-derived; the input is named.
- Computed: established by an exhaustive computer search (a computer proof); the programs are named and were run independently of each other.
- Located: the statement and its proof file were found and described; the proof was not checked.
- Checked numerically: supported by computation only; never presented as proved.

Novelty is reported only as recorded: no literature search for novelty was made except where a search is described.

Relation to the zeta reader. The reader of the split-zero programme [23] (“the zeta reader”) states the zeta programme's results and a first table of overlaps with these workbenches. This reader does not repeat them: it cites the zeta reader for the Keller-map lemma, the Hurwitz jet theorem and the Gram sandwich, and Section 5.5 lists what this audit adds to or corrects in the zeta reader's overlaps table.

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1 The Erdős–Straus workbench

The Erdős–Straus conjecture asserts that $4/n = 1/x + 1/y + 1/z$ has a solution in positive integers for every $n \geq 2$; it suffices to treat primes $n = p$. The workbench `erdos-straus-foundation` states that it claims neither a proof nor a counterexample. Its collective author name is The Clankers; the mod-107 observation behind its result R06 is credited to `u/CommonCareful3149`, further contributions to `u/UmbrellaCorp_HR`, and the diagrams behind R05 to `u/CivQ17` (README, “Attribution and collaboration”; `results/records.json`). The workbench’s fuller contribution record (`CONTRIBUTIONS.md`) also credits `u/Far-Lifeguard519` for the initiating computational investigation that prompted the project, `u/TextBackground496` and `u/JGPTech` for source ideas, and the owner (KokunoYumeto, `u/lepthymo`) for proposing the research directions and examples, directing and developing the investigations with AI, and specifying the proof and source-fidelity requirements. The results of its 22 September release are described as written proofs with Python replays, without a Lean build or independent human review. Audit note: 43_ ; first-pass report 43a_.

Throughout this section p is an odd prime, and a *solution* is a triple of positive integers with $4/p = 1/x + 1/y + 1/z$. The workbench works with $p = 12h + 1$ ($h \geq 1$).

1.1 The shell criterion

For an integer $a > p/4$ not divisible by p put $R_a = 4a - p$, $S_a = pa$ and

$$E_a = \{u : u \mid a^2, R_a \mid 4u + 1\}, \quad M_a = \{u : u \mid a^2, R_a \mid u + a\}.$$

A shell a is *occupied* if $E_a \cup M_a \neq \emptyset$.

Theorem 1.1 (shell criterion). (a) If $x \leq y \leq z$ is a solution, then $p/4 < x \leq 3p/4$. More precisely, $x < p/2$ and $x < y$, unless $p \equiv 3 \pmod{4}$ and $(x, y, z) = (\frac{p+1}{2}, \frac{p+1}{2}, \frac{p(p+1)}{4})$. For $p = 12h + 1$ this gives $3h + 1 \leq x \leq 6h$.

(b) For each integer $a > p/4$ not divisible by p (in particular each a with $p/4 < a < p$), the ordered pairs (y, z) of positive integers with $1/a + 1/y + 1/z = 4/p$ number exactly $2|E_a| + |M_a|$. A pair with $y = z$ occurs if and only if $R_a = 1$ (possible only for $p \equiv 3 \pmod{4}$ and $a = (p + 1)/4$); otherwise $|M_a|$ is even and the unordered pairs number $|E_a| + \frac{1}{2}|M_a|$. For $p \equiv 1 \pmod{4}$ this is always the case.

(c) Hence the equation is solvable at p if and only if some shell a with $p/4 < a < p/2$ is occupied.

Status. Generalised here: the workbench’s ES-01/ES-02 (`bounded_transport.tex`), audited, state it for $p = 12h + 1$ and $a \in [3h + 1, 9h]$; the proof below gives every odd prime, every $a > p/4$ with $p \nmid a$, and the halved range $x < p/2$ ($a \in [3h + 1, 6h]$ for $p = 12h + 1$). The halved range at $p \equiv 1 \pmod{12}$, with x unique, is already a reduction step of the workbench’s item 22, checked by reading in the first pass (43a_ §2.22), and items 6 and 7 work in it; the workbench’s later packages attribute these coordinates to Elsholtz and Tao [5], §2. Checked against a brute-force count of the ordered pairs on all 4,726 pairs (p, a) with $p < 260$ an odd prime and $p/4 < a < p$ (a pair with $y = z$ exactly in the 29 cases with $R_a = 1$) and on all 4,303 pairs with $p < 120$, $p/4 < a \leq 3p$ and $p \nmid a$; on all 4,398 shells of the primes $p \equiv 1 \pmod{12}$ below 700, and in the first pass on all 8,124 shells below 1000; (a) against all 2,259 solutions at the odd primes below 400; every prime $p \equiv 1 \pmod{12}$ below 30,000 has an occupied shell.

Proof. (a) The other two terms are positive, so $1/x < 4/p$; and x is the least denominator, so $4/p \leq 3/x$. Hence $p/4 < x \leq 3p/4$. Suppose $x > p/2$, so $x \geq (p + 1)/2$ as p is odd, and $x < p$. Then $2/y \geq 1/y + 1/z = 4/p - 1/x > 2/p$, so $y < p$. From $4xyz = p(xy + yz + zx)$ the prime p divides xyz , and as $0 < x \leq y < p$ it divides $z = pz'$; then $xy(4z' - 1) = pz'(x + y)$ with $p \nmid xy$, so $4z' - 1 = pk$ with $k \geq 1$. Put $N = pk + 1 = 4z'$; then $1/x + 1/y = 4k/N$, that is, $(4kx - N)(4ky - N) = N^2$ with both factors positive. But $4kx - N \geq 2k(p + 1) - N = N + 2(k - 1)$, and likewise for y ; so $k = 1$ and

$x = y = N/2 = (p+1)/2$, with $z = p(p+1)/4$, which needs $p \equiv 3 \pmod{4}$. Finally, if $x < p/2$ and $y = x$, then $1/x + 1/y > 4/p$, which is impossible. For $p = 12h + 1$ the integers in $(p/4, p/2)$ are $3h + 1, \dots, 6h$.

(b) Write $R = R_a$, $S = S_a$. Then $R \geq 1$ is odd, because p is. Since $p \nmid a$, $\gcd(a, R) = \gcd(a, p) = 1$ and $\gcd(p, R) = \gcd(p, 4a) = 1$, so S is a unit modulo R . The equation $1/y + 1/z = R/S$ is equivalent to $(Ry - S)(Rz - S) = S^2$, and both factors are positive because $1/y$ and $1/z$ are smaller than R/S . So the ordered pairs correspond bijectively to ordered factorisations $S^2 = dd'$ into positive divisors with $d \equiv d' \equiv -S \pmod{R}$, through $y = (d + S)/R$, $z = (d' + S)/R$. If $d \mid S^2$ and $d \equiv -S \pmod{R}$, then d is a unit and $d' = S^2/d \equiv S^2(-S)^{-1} = -S$; so the pairs are counted by $\#\{d \mid S^2 : d \equiv -S \pmod{R}\}$. Since $\gcd(p, a) = 1$, each divisor of $S^2 = p^2 a^2$ is uniquely $d = p^i v$ with $i \in \{0, 1, 2\}$ and $v \mid a^2$. Using $4a \equiv p \pmod{R}$: for $i = 2$, $p^2 v \equiv -pa \iff pv \equiv -a \iff 4pv \equiv -p \iff R \mid 4v + 1$, so these v form E_a ; for $i = 1$, $p v \equiv -pa \iff R \mid v + a$, so these v form M_a ; and $d \mapsto S^2/d$ maps the layer $i = 2$ bijectively onto the layer $i = 0$ while preserving the congruence. The total is $2|E_a| + |M_a|$. The swap $(y, z) \mapsto (z, y)$ is $d \mapsto S^2/d$; it maps each layer i to the layer $2 - i$, so a fixed point lies in the layer $i = 1$, where it is $v \mapsto a^2/v$ with only possible fixed point $v = a$. That needs $R \mid 2a$, i.e. $R \mid 2$ (as $\gcd(R, a) = 1$), i.e. $R = 1$. If $R \neq 1$ the swap has no fixed point, $|M_a|$ is even, $y \neq z$, and the unordered pairs number $|E_a| + \frac{1}{2}|M_a|$. For $p \equiv 1 \pmod{4}$, $R \equiv -p \equiv 3 \pmod{4}$, so $R \neq 1$.

(c) For $p \equiv 1 \pmod{4}$ it follows from (a) and (b). For $p \equiv 3 \pmod{4}$ both sides hold: the shell $a = (p+1)/4$ lies in $(p/4, p/2)$ and has $R_a = 1$, so $u = a$ lies in M_a . $\square$

The workbench also gives the exact bookkeeping around this criterion for $p = 12h + 1$: an explicit bijection between a tagged set and the ordered solutions, its inverse read off from the p -adic valuation of $Ry - S$, and the global statement that a counterexample prime would satisfy the finite, computable condition $E_a = M_a = \emptyset$ for every a (43a_§1.2; re-derived there).

1.2 The first two shells

Proposition 1.2 (the residual-3 shell). *Let $a = 3h + 1 = (p+3)/4$, so $R_a = 3$. Let $P_1 = \prod (2v_\ell(a) + 1)$ over the primes $\ell \equiv 1 \pmod{3}$ dividing a , and P_2 the same product over the primes $\ell \equiv 2 \pmod{3}$. Then $E_a = M_a = \{u \mid a^2 : u \equiv 2 \pmod{3}\}$ and $|E_a| = |M_a| = P_1(P_2 - 1)/2$. So a solution with least denominator a exists if and only if a has a prime factor $\equiv 2 \pmod{3}$, and there are then exactly $3P_1(P_2 - 1)/4$ unordered solutions with least denominator a .*

Status. Proved here; the workbench's residual-3 sieve (`first_two_shell_sieve.tex`), audited. Checked on all 4,472 primes $p \equiv 1 \pmod{12}$ below $2 \cdot 10^5$, and in the first pass on all 19,564 below 10^6 .

Proof. Since $a \equiv 1 \pmod{3}$, $3 \nmid a$. Modulo 3 both $4u + 1$ and $u + a$ are $\equiv u + 1$, so both conditions say $u \equiv 2 \pmod{3}$. A divisor $u = \prod \ell^{e_\ell}$ of a^2 has $u \equiv (-1)^{\sum e_\ell} \pmod{3}$, the sum over the primes $\ell \equiv 2 \pmod{3}$. Their exponent vectors $(0 \leq e_\ell \leq 2v_\ell(a))$ number P_2 , and the difference between the numbers with even and with odd sum is $\prod_\ell \sum_{e=0}^{2v_\ell} (-1)^e = 1$; so $(P_2 - 1)/2$ have odd sum, and the other primes contribute the factor P_1 . By Theorem 1.1(b) the unordered solutions containing a number $\frac{3}{2}P_1(P_2 - 1)/2$, and $a = 3h + 1$ is the least denominator of each of them because it is the smallest possible one. Finally $P_2 > 1$ exactly when a has a prime factor $\equiv 2 \pmod{3}$. (The count is an integer: $a \equiv 1 \pmod{3}$ makes the total exponent of those primes even, so $P_2 \equiv 1 \pmod{4}$.) $\square$

Proposition 1.3 (the residual-7 shell). *Let $a = 3h + 2 = (p+7)/4$, so $R_a = 7$ and $7 \nmid a$. Let n_3 be the total exponent in a of the primes $\equiv 3 \pmod{7}$ and n_{24} that of the primes $\equiv 2$ or $4 \pmod{7}$. The shell is occupied if and only if a has a prime factor $\equiv 5$ or $6 \pmod{7}$, or $n_3 \geq 3$, or $n_3 \geq 1$ and $n_{24} \geq 1$.*

Status. Proved here; the workbench's residual-7 sieve, audited, together with its count as coefficients of $\prod_\ell \sum_{e=0}^{2v_\ell(a)} T^{\lambda(\ell)e}$ in $\mathbb{Z}[T]/(T^6 - 1)$. Checked on all 4,472 primes $p \equiv 1 \pmod{12}$ below $2 \cdot 10^5$ (and below 10^6 in the first pass).

Proof. The exterior condition $7 \mid 4u + 1$ is $u \equiv -4^{-1} \equiv 5 \equiv 3^5$, and the middle condition is $u/a \equiv -1 \equiv 3^3 \pmod{7}$; in base 3 the discrete logarithms of $1, \dots, 6$ are $0, 2, 1, 4, 5, 3$. *Sufficiency.* A prime $\ell \equiv 5$ dividing a gives $u = \ell$; a prime $\ell \equiv 6$ gives $u = a\ell$. If $n_3 \geq 3$, a product of primes $\equiv 3$ dividing a^2 with total exponent 5 gives $u \equiv 3^5$. If primes $t \equiv 3$ and $\ell \in \{2, 4\} \pmod{7}$ divide a , then $u = alt$ (when $\ell \equiv 2 \equiv 3^2$) or $u = a\ell/t$ (when $\ell \equiv 4 \equiv 3^4$) divides a^2 and has $u/a \equiv 3^3$. *Necessity.* If none of the conditions holds, every prime factor of a is $\equiv 1, 2, 3$ or 4 , and either $n_3 = 0$ or $(1 \leq n_3 \leq 2 \text{ and } n_{24} = 0)$. If $n_3 = 0$, a and all divisors of a^2 lie in the subgroup $\{1, 2, 4\}$ of squares, which contains neither 5 nor -1 . Otherwise $a \equiv 3^{n_3}$ and a divisor of a^2 is $\equiv 3^e$ with $0 \leq e \leq 2n_3 \leq 4$; the exterior target needs $e \equiv 5 \pmod{6}$ and the middle one $e - n_3 \equiv 3 \pmod{6}$ with $|e - n_3| \leq 2$, both impossible. $\square$

The shells with $R = 3$ and $R = 7$ exist exactly for the primes $p \equiv 1 \pmod{4}$, with $a = (p+3)/4$ and $a = (p+7)/4$ (Generalised here; checked on all 1,611 such primes below $3 \cdot 10^4$). Proposition 1.3 holds for all of them, since its proof never uses $p \equiv 1 \pmod{3}$. Proposition 1.2 holds as stated for $p \equiv 1 \pmod{12}$ only. For $p \equiv 5 \pmod{12}$, $a = (p+3)/4 \equiv 2 \pmod{3}$, so modulo 3 the middle condition reads $u \equiv 1$; then $E_a = \{u \mid a^2 : u \equiv 2\}$ and $M_a = \{u \mid a^2 : u \equiv 1\}$, with $|E_a| = P_1(P_2 - 1)/2$ and $|M_a| = P_1(P_2 + 1)/2$ by the parity count of the proof above, and there are exactly $|E_a| + \frac{1}{2}|M_a| = P_1(3P_2 - 1)/4$ unordered solutions with least denominator a (an integer: the total exponent in a of the primes $\equiv 2 \pmod{3}$ is odd, so $P_2 \equiv 3 \pmod{4}$). That shell is always occupied ($1 \in M_a$), and its occupancy criterion holds trivially, since $a \equiv 2 \pmod{3}$ has a prime factor $\equiv 2 \pmod{3}$; those primes are solvable anyway, since $p \equiv 2 \pmod{3}$.

Proposition 1.4 (the two-shell survivors). *Let $p \equiv 1 \pmod{12}$ be prime. Both shells $a = (p+3)/4$ and $a = (p+7)/4$ are unoccupied if and only if $p \equiv 1 \pmod{24}$, every prime factor of $(p+3)/4$ is $\equiv 1 \pmod{3}$, and every prime factor of $(p+7)/4$ is $\equiv 1, 2$ or $4 \pmod{7}$. Every such p is a square modulo 7, so $p \equiv 1, 25$ or $121 \pmod{168}$.*

Status. Proved here; the first pass stated the congruence mod 24 as an observation on its computation. Checked on all primes $p \equiv 1 \pmod{12}$ below $2 \cdot 10^5$.

Proof. If $p \equiv 13 \pmod{24}$, then $(p+3)/4$ is even and $2 \equiv 2 \pmod{3}$, so the first shell is occupied (Proposition 1.2). If $p \equiv 1 \pmod{24}$, the first shell is unoccupied exactly when every prime factor of $(p+3)/4$ is $\equiv 1 \pmod{3}$; and $(p+7)/4$ is even, so $n_{24} \geq 1$ and Proposition 1.3 says that the second shell is occupied exactly when $(p+7)/4$ has a prime factor $\equiv 3, 5$ or $6 \pmod{7}$, a non-residue. If it has none, $(p+7)/4$ is a square modulo 7, and so is $p \equiv 4 \cdot \frac{p+7}{4}$. Among the classes $\equiv 1 \pmod{24}$ modulo 168, the squares modulo 7 are 1, 25 and 121. $\square$

Of the primes $p \equiv 1 \pmod{12}$ below 10^6 , exactly 989 have both shells unoccupied, the first being 1129, 1201, 2521 (first pass).

1.3 Which primes remain

The workbench proves solvability for $p \equiv 2 \pmod{3}$, $p \equiv 3 \pmod{4}$ and $p \equiv 13 \pmod{24}$ (scope identities that it does not advertise as new) by the families $(p, \frac{p+1}{3}, \frac{p(p+1)}{3})$, $(\frac{p+1}{4}, \frac{p(p+1)}{2}, \frac{p(p+1)}{2})$ and $(a, \frac{pa}{2}, pa)$ with $a = \frac{p+3}{4}$ (even when $p \equiv 13 \pmod{24}$); so its own reduction leaves the primes $p \equiv 1 \pmod{24}$. Its later packages restrict to the *hard* primes $p \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$, which are exactly the unit squares modulo 840 (a group of order $1 \cdot 1 \cdot 2 \cdot 3 = 6$). The reduction to them is classical (Mordell [15]) and is not proved in the texts read; five identities prove it.

Proposition 1.5 (the reduction modulo 840). *Let $p \equiv 1 \pmod{24}$ be prime, and let (A, B, R) be $(1, 2, 7)$, $(2, 1, 7)$, $(1, 1, 7)$, $(1, 2, 15)$ or $(2, 1, 15)$ according as $p \equiv 3, 5, 6 \pmod{7}$ or $p \equiv 2, 3 \pmod{5}$.*

Then $D = (p + R)/(4AB)$ and $C = (A + pB)/R$ are positive integers and

$$\frac{4}{p} = \frac{1}{ABD} + \frac{1}{ACD} + \frac{1}{pBCD}.$$

So the equation is solvable at every prime $p \equiv 1 \pmod{24}$ that is not a square modulo 840.

Status. Proved here (the reduction is classical); the first pass covered the 18 non-square unit classes $\equiv 1 \pmod{24}$ by the same kind of families. Checked on all primes $p \equiv 1 \pmod{24}$ below $3 \cdot 10^5$.

Proof. With $RC = A + pB$ and $4ABD = p + R$: $\frac{1}{ABD} + \frac{1}{ACD} + \frac{1}{pBCD} = \frac{pC + pB + A}{pABCD} = \frac{C(p+R)}{pABCD} = \frac{p+R}{pABD} = \frac{4}{p}$. Since $p \equiv 1 \pmod{8}$, 8 divides $p + 7$ and $p + 15$, so $4AB \mid p + R$; $R \mid A + pB$ is the stated congruence modulo 7 or 5, together with $A + pB \equiv A + B \equiv 0 \pmod{3}$ when $R = 15$. The primes $p \equiv 1 \pmod{24}$ left over have $p \equiv 1, 2, 4 \pmod{7}$ and $p \equiv \pm 1 \pmod{5}$, which are exactly the unit squares modulo 840. $\square$

1.4 The structural items

The workbench's 28 structural items (17–23 September) are listed in 43a_§2. Those audited:

- *Item 6* (audited). With $L_a = 8\tau(a)^2/\varphi(R_a) - \mathcal{E}_a$, where $\mathcal{E}_a$ is the residue-collision energy of the $2\tau(a)$ divisors of pa modulo R_a : for $p \equiv 1 \pmod{4}$, $p \geq 2^{20}$, $\sum_{a \in (p/4, p/2)} L_a \leq -\frac{1}{24}p \log p$. So the lower bound that keeps only the principal character, summed over the shells, is negative for large p : that method cannot prove occupancy.
- *Item 7* (audited). For $p \equiv 1 \pmod{8}$, bounds in terms of the least quadratic nonresidue; at the hard primes a middle state (a, u) on a shell $p/4 < a < p/2$ has $a \leq 3(p + 3)/11$, and an exterior state satisfies $22(p + 1) \leq 23(p - 2a + 1)^2$. The workbench says this localises solutions and does not prove the region occupied.
- *Item 14* (audited). For $h' = 4j - 1 > t$ the divisors $W \mid j^2$ with $W \equiv t \pmod{h'}$ are canonical, companion or finite templates; the finite ones satisfy the sharp bound $h' \leq t^2 - 3t + 1$, with equality exactly in one parametrised family, attained at infinitely many primes $p \equiv 1 \pmod{840}$. For every $t \geq 4$ infinitely many hard primes have an exterior state whose same-grade middle box is empty.
- *Item 21* (audited). For a proper rational trace source the complete set of same-word residual returns is $\{k \mid R : d \mid k, k \equiv 1 \pmod{4K(u)}\}$; a uniform return exists exactly for the nine words $u \mid 36$.
- *Item 22* (audited, certificate). For every solution at $p \equiv 1 \pmod{12}$ an explicit symmetric polynomial $\mathfrak{N}(p, x, y, z)$ of degree 8 satisfies $\mathfrak{N} < -\frac{125873811}{262144}p^8$ ($\approx -480.17p^8$), and $\mathfrak{N} < -6960.19p^8$ when $p \equiv 1 \pmod{24}$; the proof reduces to finite polynomial positivity certificates, recomputed independently. It is a statement about existing solutions, not about their existence.
- *Item 26* (audited in part: the least example and the deduction from it; its prime-square trace theorem at $p \equiv 1 \pmod{24}$ and its terminality statement read and correct as read, with the identity $16(4q^3 + 1) - (4q + 1)(16q^2 - 4q + 1) = 15$ checked; the factor-sum, Pell and fixed-tail-sum sections not checked). $p = 67369$ is the least prime $p \equiv 1 \pmod{24}$ whose first “trace shell” ($a = 16849, R = 27$) precedes its first full shell ($a = 16850, R = 31$); so no universal map from rational traces to solutions avoids increasing a .
- *Item 4* (conditional on a computer enumeration of 1,381,117,764 rooted profiles, 5,922,259 of them range-compatible, run by two implementations of the workbench and not re-run here). At every hard prime, from any nonresidue prime, factoring at most 25 numbers $(p + \sigma qt^2)/4$ with $t \in \{1, 3, 5, 7, 9\}$ yields at least six distinct nonresidue primes. A direct test of all 108 hard primes below 40,000 agrees.

Items 2, 3 and 8 are audited (certificate); the 22-digit prime 7510085481569082811681 of item 3 is proved prime here by a Pocklington chain ($p - 1 = 2^5 \cdot 3 \cdot 5 \cdot q_1$, $q_1 - 1 = 2 \cdot 3 \cdot 5 \cdot 351707 \cdot q_2$, $q_2 - 1 = 2 \cdot 3 \cdot 17^2 \cdot 5051 \cdot 169307$, witnesses below 12 at every level, the leaves by trial division). Item 23 is audited in part (its certificates; the deduction read in part); items 11–13, 15 and 20 use imported material (Section 5); the remaining 17 items, including these five, are located. The results bench R01–R10 and the arguments X1–X7 are checked in 43a_§3; one source passage error ($X5: 289^3 \equiv 169$, not -1 , modulo 840) is recorded by the workbench itself.

1.5 The $(3, 4, \infty)$ monodromy covers

A package of the workbench (ES-09) imports three integer matrices A_1, A_2, A_∞ with $A_1^3 = A_2^4 = A_1 A_2 A_\infty = I$ from the S^6 manuscript [1] and nothing else from it. For odd levels D , which is the source's standing hypothesis, the induced affine action on $(\mathbb{Z}/D)^3$ is transitive when $3 \nmid D$ and has two orbits, of sizes $D^3/9$ and $8D^3/9$, when $3 \mid D$; for $3 \nmid D$ the cover is connected of genus $(5D^3 - 12D^2 - 17D + 24)/24$, and for $3 \mid D$ the two orbits give components of genera $(5D^3 - 12D^2 - 81D + 216)/216$ and $(5D^3 - 12D^2 - 9D + 27)/27$. These statements are false for even D (at $D = 2$ the orbits have sizes 2 and 6; found by the referee of the notes), which the ES application never uses, since there D divides the odd number R_a . The package also shows that the equation fails at p if and only if a denominator-labelled cover has no section. That equivalence is correct, and it is Theorem 1.1 again: a section exists exactly when the tail denominator $R_a / \gcd(R_a, 4u + 1)$ or $R_a / \gcd(R_a, u + a)$ equals 1.

Status. Audited: the proof read; the relations, the orbits and the genera also recomputed from cycle counts for $D = 1, 3, \dots, 15$. The equivalence is a re-encoding of Theorem 1.1; the assessment is this audit's (the package presents it as a complete failure object).

1.6 Negative results, with exact scope

Each row says which proposed implication fails and what the failure does not show. Every certificate prime in the table is itself solvable.

Source	What fails	Certificate	What it does not show
bounded transport	every occupied shell has a solution with unary layer $A \leq 2$	$p = 37, a = 12$: only $u = 8$	the global cutoff ($a = 10$ works at 37)
first-two-shell sieve	occupancy of the $R = 7$ shell gives a first-two-layer hit	$p = 373, a = 95$	anything about other shells
item 2	every canonical closed component contains a local hit	$p = 2521$: a 7-cycle sink	2521 is solvable
item 3	strict canonical descent ends at a hit	a closed triangle at a 22-digit prime	that prime is solvable
item 6	the principal-only bound, summed over shells, is positive for large p	$\sum L_a \leq -\frac{1}{24}p \log p$	anything about the true counts
item 6	a nonnegative reweighting of it certifies occupancy	$p = 87481$: all 21,870 shells have $L_a \leq 0$	only this certificate family fails
item 14	same-grade middle repair from exterior states with $\alpha = -4t, t \geq 4$	$p = 7840561$	those primes are solvable
item 21	a same-word residual return for words $u \nmid 36$	3,041 of the 3,654 sources with $u \nmid 36$ at primes below 2600	endpoint solutions exist
item 26	a trace-to-solution map never increases a	$p = 67369$	67369 is solvable

All of these are statements about methods or certificate families. None is a counterexample to the conjecture, and none bears on whether it is true.

2 The Collatz workbench

The workbench `collatz-workbench` states in every note that it does not claim the Collatz conjecture; its chapters describe themselves as written proofs with exact finite checks, not externally reviewed. Its public collaboration name is Kokuno Yumeto, and its contributions were developed with ChatGPT 5.6 Sol (in Codex) and GPT-6 Astra (README). An early tree of the workbench is quarantined by the owner's directive; the quarantine withdraws those statements from use and does not show any of them false. Audit note: 44₋; first-pass report 44a₋.

Notation. X is the set of positive odd integers and $T(n) = (3n + 1)/2^{\nu_2(3n+1)}$ on X . The exponent word of n of length m is $(a_1, \dots, a_m)$ with $a_i = \nu_2(3x_{i-1} + 1)$, $x_0 = n$, $x_i = T(x_{i-1})$; $A = a_1 + \dots + a_m$.

2.1 History laws

Proposition 2.1 (cylinders). *For every exponent word w of length m and sum A there is a residue h_w modulo 2^A such that an odd $n = 2k + 1$ has exponent word w exactly when $k \equiv h_w \pmod{2^A}$. Among the N consecutive odd starts $2k + 1$, $b \leq k \leq b + N - 1$, the number with word w is $\lfloor (b + N - 1 - h_w)/2^A \rfloor - \lfloor (b - 1 - h_w)/2^A \rfloor$, which is $\lfloor N/2^A \rfloor$ or $\lceil N/2^A \rceil$; so the empirical frequency $P(w)$ satisfies $|P(w) - 2^{-A}| \leq (1 - 2^{-A})/N < 1/N$.*

Status. Audited; classical (Terras [22], Everett [7]); the workbench credits the mechanism to Tao's Proposition 1.9 [20] and claims no priority.

Theorem 2.2 (the full-history threshold). *Let P be the empirical law of the first m exponents over the N starts of Proposition 2.1, $G_m(w) = 2^{-A(w)}$ (independent exponents with $\Pr(a = j) = 2^{-j}$), $L = \log_2 N$, $\Delta = \text{TV}(P, G_m)$, and Φ the standard normal distribution function.*

- (a) *For every $H \geq 0$, $\max\{0, Q_m(H) - N2^{-H-1}\} \leq \Delta \leq \min\{1, Q_m(H) + \binom{H}{m}/N\}$, where $Q_m(H) = \Pr(\text{Bin}(H, \frac{1}{2}) \leq m - 1)$ (so $Q_m(H) = 1$ and $\binom{H}{m} = 0$ for $H < m$).*
- (b) *If $m = \lfloor L/2 + c\sqrt{L} \rfloor$, then $\sup_b |\Delta - \Phi(2c)| \rightarrow 0$ as $N \rightarrow \infty$.*
- (c) *If $0 < \delta < \frac{1}{2}$ and $1 \leq m \leq (\frac{1}{2} - \delta)L$, then $\Delta \leq N^{-\delta} + N^{-\delta^2/(2\ln 2)}$. If $m \geq (\frac{1}{2} + \delta)L$ and $\delta L > 1$, then, with $H = \lceil (1 + \delta)L \rceil$, $1 - \Delta \leq \frac{1}{2}N^{-\delta} + \exp(-2(\frac{\delta L}{2} - \frac{1}{2})^2/H)$. So $\Delta \rightarrow 0$, respectively $\Delta \rightarrow 1$, at a polynomial rate in N .*

All statements are uniform in b .

Status. Proved here; the workbench's history-law note, audited, which states the uniformity in b and gives (c) as its explicit estimates (4.3)–(4.4) (the first “for sufficiently large N ”; the proof below needs no such condition). Checked: 2,952 sandwich checks in the first pass; here the exact Δ for $N = 2^{10}, 2^{12}, 2^{14}$ at three offsets b against (a) for every $0 \leq H \leq 2L + 2$ and against (c); at $N = 2^{20}$ the TV values are 0.004, 0.044, 0.342, 0.668, 0.895 for $c = -1, -\frac{1}{2}, 0, \frac{1}{2}, 1$, against $\Phi(2c) = 0.023, 0.159, 0.5, 0.841, 0.977$ (slow convergence, as the note warns). No literature search on the threshold was made.

Proof. Under G_m the sum A is the waiting time for the m -th success in fair coin flips, so $G_m(A > H) = Q_m(H)$, and the words with $A \leq H$ number $\binom{H}{m}$. Upper bound: $\Delta = 1 - \sum_w \min(P(w), G(w))$, and by Proposition 2.1 the words with $A \leq H$ contribute at least $\sum_{A \leq H} (2^{-A} - 1/N) = 1 - Q_m(H) - \binom{H}{m}/N$. Lower bound: P is carried by at most N

words, each word with $A > H$ has G -mass at most 2^{-H-1} , so $\sum_{A>H} \min(P, G) \leq N2^{-H-1}$, while $\sum_{A \leq H} \min(P, G) \leq 1 - Q_m(H)$. Nothing here needs $H \geq m$: for $H < m$ no word has $A \leq H$, and every word has G -mass at most $2^{-m} \leq 2^{-H-1}$. (b) Take $H_+ = L + L^{1/4}$ in the lower bound and $H_- = L - L^{1/4}$ in the upper bound: $N2^{-H_+-1} \rightarrow 0$ and $(\frac{H_-}{m})/N \leq 2^{H_- - L} \rightarrow 0$. By the central limit theorem for $\text{Bin}(H, \frac{1}{2})$, uniform in the argument with error $O(H^{-1/2})$, $Q_m(H_\pm) = \Phi((2m - 2 - H_\pm)/\sqrt{H_\pm}) + o(1)$, and $(2m - 2 - H_\pm)/\sqrt{H_\pm} \rightarrow 2c$. (c) Let $X \sim \text{Bin}(H, \frac{1}{2})$; Hoeffding's inequality gives $\Pr(X \leq \frac{H}{2} - u)$ and $\Pr(X \geq \frac{H}{2} + u)$ at most $e^{-2u^2/H}$ for $u \geq 0$. First let $m \leq (\frac{1}{2} - \delta)L$ and $H = \lfloor (1 - \delta)L \rfloor$, which is at least 1 since $L \geq 1/(\frac{1}{2} - \delta) > 2$. Then $(\frac{H}{m})/N \leq 2^H/N \leq N^{-\delta}$, and $u = \frac{H}{2} - (m - 1) \geq \frac{\delta L}{2} + \frac{1}{2}$, so $Q_m(H) \leq e^{-2(\delta L/2)^2/H} \leq e^{-\delta^2 L/2} = N^{-\delta^2/(2 \ln 2)}$ as $H \leq L$. Next let $m \geq (\frac{1}{2} + \delta)L$, $\delta L > 1$ and $H = \lceil (1 + \delta)L \rceil$. Then $N2^{-H-1} \leq \frac{1}{2}N^{-\delta}$, and $1 - Q_m(H) = \Pr(X \geq m)$, which is 0 if $m > H$ and otherwise at most $e^{-2u^2/H}$ with $u = m - \frac{H}{2} \geq \frac{\delta L}{2} - \frac{1}{2} > 0$. Both cases follow from (a), and neither bound involves b . $\square$

Part (c) is an obstruction: above $(\frac{1}{2} + \delta) \log_2 N$ halving counts, the whole word law of N consecutive starts is far from the geometric model. It says nothing about coarser observables or about individual orbits.

2.2 Merging families

Lemma 2.3. *For even $e \geq 2$, $h_e = (2^e + 2)/3$ is an integer with $h_e \equiv 2 \pmod{4}$ and $v_3(h_e) = v_3(e - 1)$.*

Status. Proved here; the workbench's Lemma 1 of the all-even join extension, audited; checked for all even $e \leq 3000$.

Proof. $2^e \equiv 1 \pmod{3}$ gives integrality; $2^e + 2 \equiv 2 \pmod{4}$ and 3 is odd. With $k = e - 1$ odd, $2^e + 2 = 2(2^k + 1)$ and the lifting-the-exponent lemma gives $v_3(2^k + 1) = v_3(3) + v_3(k)$, so $v_3(h_e) = v_3(k)$. $\square$

Theorem 2.4 (explicit merging families). *Fix an even $e \geq 2$, $b \geq 1$ and $\sigma \in \{0, 1\}$. Let a be the least $a \geq 1$ with $2^{a+e+2b-\sigma} < 3^{a+b}$, and put $J = 2^{e+2b-\sigma}$, $Q = 3^{a+b}$, $K = 2^a J$, $(d, r) = (4, 3)$ if $\sigma = 0$ and $(32, 27)$ if $\sigma = 1$. Let $n_0 \in (0, dQ)$ solve $n_0 \equiv r \pmod{d}$ and $Jn_0 + h_e 3^b \equiv 0 \pmod{Q}$, and $m_0 = (Kn_0 + 2^a h_e 3^b)/Q - 1$. For every $v \geq 0$, $n = n_0 + dQv$ and $m = m_0 + dKv$ are odd, $0 < m < n$, $v_3(n) = b + v_3(e - 1)$, and the exponent words of n and m begin with (1) and $(1^a, e, 2^{b-1}, 3)$ if $\sigma = 0$, and with $(1, 2, 1)$ and $(1^a, e, 2^{b-1}, 1, 1, 3)$ if $\sigma = 1$, and end at the same odd number.*

Status. Proved here; the workbench's Theorem 2 of the all-even join extension (16 September), audited, together with its converse (the endpoint equality forces this progression). Checked from the definitions on 600 pairs ($e \leq 20$, $b \leq 6$, both σ , including $v = 10^{12} + 3$), and by the referee of the notes on 1,200 further pairs.

Proof. Range of a . At $a = b + 2e - 2\sigma$, $K/Q = (8/9)^{b+e-\sigma} < 1$, so $a \leq b + 2e - 2\sigma$; and $K < Q$ needs $(3/2)^a > 2^{e-\sigma}(4/3)^b \geq 2^{e-1}$, so $a \geq e > v_3(e - 1)$. The 3-adic part. J is a unit mod 3, so $n \equiv -h_e 3^b J^{-1} \pmod{3^{a+b}}$, and since $v_3(h_e 3^b) = b + v_3(e - 1) < a + b$, $v_3(n) = b + v_3(e - 1)$. Put $z = n/3^b$ and $W = (Jz + h_e)/3^a \in \mathbb{Z}$; as $4 \mid J$ and $h_e \equiv 2 \pmod{4}$, $v_2(W) = 1$, and $m = 2^a W - 1$. The right path. Since $m + 1 = 2^{a+1}(W/2)$ with $W/2$ odd, the first a steps from m have exponent 1, through $x_j = 3^j 2^{a-j} W - 1$, ending at $x_a = 3^a W - 1 = Jz + h_e - 1$, which is odd. By $3h_e - 2 = 2^e$, $3x_a + 1 = 2^e(3 \cdot 2^{2b-\sigma} z + 1)$ with $2b - \sigma \geq 1$, so the next exponent is exactly e . The next $b - 1$ steps have exponent 2 (the factor $3 \cdot 2^{2b-\sigma}$ loses a factor 4 and gains a 3 at each step) and end at $4n + 1$ ($\sigma = 0$) or $2n + 1$ ($\sigma = 1$). The meeting point. If $\sigma = 0$, $n \equiv 3 \pmod{4}$, so $T(n) = (3n + 1)/2$

with exponent 1, and $3(4n + 1) + 1 = 4(3n + 1) = 8 \cdot \frac{3n+1}{2}$ gives exponent 3 and the same value. If $\sigma = 1$, $n \equiv 27 \pmod{32}$ gives $n \rightarrow \frac{3n+1}{2} \rightarrow \frac{9n+5}{8} \rightarrow \frac{27n+23}{16}$ with exponents 1, 2, 1, and $2n+1 \rightarrow 3n+2 \rightarrow \frac{9n+7}{2} \rightarrow \frac{27n+23}{16}$ with exponents 1, 1, 3. *Height.* $m = \frac{K}{Q}n + h_e(\frac{2}{3})^a - 1$ with $K < Q$, and $h_e(\frac{2}{3})^a < \frac{h_e 3^b}{J} = \frac{2+2^{2-e}}{3} 2^{\sigma-1}(\frac{3}{4})^b < 1$. $\square$

For example ($e = 8, b = 2, \sigma = 0, a = 16$): $n = 20,241,207 + 1,549,681,956v$ and $m = 14,024,703 + 2^{30}v$ reach $30,361,811 + 2,324,522,934v$ for every $v \geq 0$. The workbench also proves (its Theorem 3; audited) that whether an odd $n > 1$ is the larger member of such a pair is decided by a finite search at n : $e \leq \lfloor \log_2(n-1) \rfloor - 1$, $b = \nu_3(n) - \nu_3(e-1)$, and the admitted smaller members are $m_{a'} = 2^{a'}(J(n/3^b) + h_e)/3^{a'} - 1$ for $a \leq a' \leq \nu_3(J(n/3^b) + h_e)$. Only 367 of the odd $n < 400,000$ admit a template. The workbench notes that the search does not show that a template applies to every retained source; no convergence statement follows.

2.3 Descent, and a contraction in a model

Proposition 2.5. *Let $F_m(x) = (3^m x + C_m)/2^{A_m}$ be the affine map of the first m odd steps of an exponent word ($C_m \geq 1$), and n_0 the least odd $n \geq 3$ whose word begins with this one (its cylinder is a residue class modulo 2^{A_m+1}). Then every odd $n \geq 3$ whose word begins with this one satisfies $T^m(n) < n$ if and only if $F_m(n_0) < n_0$. In particular this holds if $F_m(3) < 3$. More precisely, a member n satisfies $T^m(n) < n$ if and only if $D = 2^{A_m} - 3^m > 0$ and $n > C_m/D$.*

Status. Generalised here: the workbench's Chapter 2, Theorem 2.1 (audited; 1,601 leaves and 9,605 starts checked in the first pass) assumes $F_j(3) \geq 3$ for $j < m$ and $F_m(3) < 3$, which makes the word a first stopping leaf; the descent needs only $F_m(n_0) < n_0$, and that condition is also necessary. Checked on 36,267 cases with $n < 2^{14}$, $m \leq 8$ (the condition $F_m(3) < 3$), and on all 637 words with $m \leq 5$ and $A_m \leq 10$ against the actual descent of every member below 2^{15} (the equivalence). For example, the word $(1, 1, 1, 1, 4)$ has $F_5(3) = 235/64 > 3$, but $n_0 = 95$ and every member descends.

Proof. With $D = 2^{A_m} - 3^m$ and $C = C_m$, $F_m(x) - x = (C - Dx)/2^{A_m}$, and $T^m(n) = F_m(n)$ for every member n ; so a member descends exactly when $Dn > C$, which, as $C > 0 < n$, means $D > 0$ and $n > C/D$. If $F_m(n_0) < n_0$, then $C < Dn_0$, so $D > 0$ (as $C > 0$), and every member $n \geq 3$ has $n \geq n_0$ and $Dn \geq Dn_0 > C$. The converse is the case $n = n_0$. If $F_m(3) < 3$, then $C < 3D$, so $D > 0$ and $Dn_0 \geq 3D > C$. $\square$

Proposition 2.6. *For $x \geq 2$, $\frac{1}{2}(\sqrt{(x-2)/2} + \sqrt{(3x-1)/2}) \leq \frac{\sqrt{15}}{4}\sqrt{x-1}$, with equality exactly at $x = 7$.*

Status. Generalised here: the workbench's Chapter 2, Theorem 7.1 (audited) states it for $x \geq 3$; the proof gives $x \geq 2$.

Proof. With $u = (x-2)/2$, $v = (3x-1)/2$: $\sqrt{u} + \sqrt{v} = \sqrt{2u}\sqrt{1/2} + \sqrt{v} \cdot 1 \leq \sqrt{2u+v}\sqrt{3/2}$ and $2u+v = \frac{5}{2}(x-1)$. Equality needs $4u = v$, that is $x = 7$. $\square$

In the fair-coin model, where x moves to $x/2$ or $(3x+1)/2$ with probability $\frac{1}{2}$ each, this says that $\mathbb{E}\sqrt{X_{\text{next}}-1} \leq \frac{\sqrt{15}}{4}\sqrt{x-1}$, also when branches that fall below 3 are killed. Started at $X_0 = 3$ with the first step odd ($X_1 = 5$), and with τ the first time below 3, iteration gives $\Pr(\tau > h) \leq \mathbb{E}[\sqrt{X_h-1}; \tau > h]/\sqrt{2} \leq \sqrt{2}(\frac{\sqrt{15}}{4})^{h-1} \leq \frac{3}{2}(\frac{31}{32})^h$ for $h \geq 1$, so the killed process dies almost surely; and the completed stopping word, whose tail t_H satisfies $t_0 = 1$ and $t_H = \frac{1}{2}(t_{H-1} + \Pr(\tau > H))$, has $t_H \leq \frac{31}{20}(\frac{31}{32})^H$ by induction, since $\frac{1}{2}(\frac{31}{20} \cdot \frac{32}{31} + \frac{3}{2}) = \frac{31}{20}$ (the workbench's (7.9)–(7.16), audited in 44a_§2.4). These are statements about the model, not about orbits, as the workbench says.

2.4 Cycles

Proposition 2.7. *No nontrivial positive cycle of T has at most one step with exponent 1.*

Status. Proved here; the workbench's Chapter 3, audited. Weaker than the published cycle results, as the chapter says.

Proof. Let $n > 1$ be the least element of the cycle. A step from $x > 1$ with exponent at least 2 gives $T(x) \leq (3x + 1)/4 < x$; so some step has exponent 1, and if only one does, it is the step from n . Then $x_1 = (3n + 1)/2$ and every later step decreases; a later exponent ≥ 3 would give a value at most $(3x_1 + 1)/8 = (9n + 5)/16 < n$. So the word is $(1, 2, \dots, 2)$ with $m - 1$ twos, and $x_{j+1} - 1 = \frac{3}{4}(x_j - 1)$ along the twos. Closing the cycle, $n - 1 = (\frac{3}{4})^{m-1} \frac{3n-1}{2}$, i.e. $nD = 2 \cdot 4^{m-1} - 3^{m-1}$ with $D = 2 \cdot 4^{m-1} - 3^m$, and $n - 1 = 2 \cdot 3^{m-1}/D$. D is odd and prime to 3, so $D = \pm 1$; but $D = -1$ for $m = 1, 2$ (the negative cycles -1 and -5) and $D_{m+1} = 4D_m + 3^m \geq 5$ for $m \geq 2$. $\square$

Theorem 2.8. *Every nontrivial positive cycle of T has at least 1636 odd steps, at least 2593 halvings, and more than 678 steps with exponent 1.*

Status. Proved here; the workbench's Chapter 6, Theorem 9, audited, states the bound on steps with exponent 1; the bounds on m and A come out of the same proof. Its hypothesis, that every odd $n \leq 330,749$ reaches 1 (the workbench's Theorem 8, from its own database), was recomputed for this reader: every odd n in that range falls below itself under T .

Proof. Let the cycle have m odd steps, k of them with exponent 1, total exponent A and least element s ; since no element of the cycle reaches 1, $s \geq 330,751$. Multiplying $T(x_i)/x_i = (3+1/x_i)/2^{a_i}$ around the cycle gives $2^A = \prod (3 + 1/x_i) \in (3^m, (3 + 1/s)^m]$, and $A \geq 2m - k$ gives $(4s)^m \leq 2^k(3s + 1)^m$. The least m for which a power of 2 lies in $(3^m, (3 + 1/s)^m]$ is $m = 1636$ (exact integer arithmetic), so $A > 1636 \log_2 3 = 2592.998 \dots$; and $(4s/(3s + 1))^m$ increases with m ; at $m = 1636$ the inequality needs $k \geq 679$. $\square$

The count k differs from the number of local minima used in the published cycle bounds (Simons-de Weger [19]; Hercher [11], no m -cycles for $m \leq 91$), and the workbench labels the result internal. The argument is Eliahou's [10]: a cycle whose elements all exceed s has $2^A \in (3^m, (3 + 1/s)^m]$, so A/m lies in $(\log_2 3, \log_2(3 + 1/s)]$ and m is at least the least denominator of a fraction in that interval. Run with a larger verification range, it gives far more. A verification made for this reader (descent_2e33.c; 38 seconds on two cores, 128-bit arithmetic, largest value met $9.07 \cdot 10^{18}$) shows that every odd n with $3 \leq n < 2^{33}$ falls below itself under $n \mapsto (3n + 1)/2$, $n \mapsto n/2$; so every $n < 2^{33}$ reaches 1, and with no external input the same argument gives $m \geq 190,537$, $A \geq 301,994$ and $k \geq 79,080$ (the least denominator is that of $301994/190537$; Computed: the verification was repeated by two programs written independently by the referees of this reader, both with the odd-to-odd map). A published verification range gives more still.

Corollary 2.9. *Assume that every $n < 2.17 \cdot 10^{20}$ reaches 1. This holds: Barina verified every $n < 2^{68}$ (2020, published in 2021 [2]) and every $n < 2^{71}$ [3]. Then every nontrivial positive cycle of T has at least 72,057,431,991 odd steps, at least 114,208,327,604 halvings (so at least 186,265,759,595 steps of $n \mapsto n/2$, $n \mapsto 3n + 1$), and at least 29,906,536,378 steps with exponent 1.*

Status. Conditional on Barina's published computation, and its 2021 range 2^{68} already suffices (the 2025 paper, §1, attributes that range to the 2021 paper; the project page dates it 7 May 2020). The derivation is proved here (and is Eliahou's method). The fraction $114208327604/72057431991$ is the fraction of least denominator in $(\log_2 3, \log_2(3 + 1/s)]$ exactly for $s_{lo} < s \leq s_{hi}$, with $s_{lo} = 1/(2^{10439860591/6586818670} - 3) = 2.1689015534 \cdot 10^{20} \approx 2^{67.556}$ (from the previous best upper approximation of $\log_2 3$) and $s_{hi} = 1/(2^{114208327604/72057431991} - 3) = 4.3585 \cdot 10^{21} \approx 2^{71.884}$; so, by this argument, any verified range

between $2^{67.56}$ and $2^{71.88}$ gives the same bounds. The published bound, stated after the proof, is stronger. The fraction was found by the continued-fraction algorithm for the simplest fraction in an interval (80-digit arithmetic) and confirmed by an independent Stern–Brocot descent (120-digit arithmetic). The same computation returns $m = 1636$, $k \geq 679$ at $s = 330,751$ (Theorem 2.8) and $A = 17,087,915$ at $s = 2^{40}$, which is Eliahou’s published bound [10].

Proof. The least element s of a nontrivial cycle exceeds s_{10} . As in the proof of Theorem 2.8, $A/m \in (\log_2 3, \log_2(3 + 1/s)] \subseteq (\log_2 3, \log_2(3 + 1/s_{10}))$. That interval excludes $10439860591/6586818670$, and every fraction in $(\log_2 3, 10439860591/6586818670)$ has denominator at least $72,057,431,991$, the denominator of the next best upper approximation; so $m \geq 72,057,431,991$. Then $A > m \log_2 3 \geq 72,057,431,991 \log_2 3 = 114,208,327,603.9999999999 \dots$, so $A \geq 114,208,327,604$; and $k \geq m \log_2(4s/(3s + 1)) \geq 72,057,431,991 \cdot (2 - \log_2(3 + 1/s_{10})) = 29,906,536,377.99 \dots$ $\square$

Corollary 2.9 records what Eliahou’s argument alone gives from Barina’s 2021 range; the published bound is stronger. Hercher [11, Corollary 29] proves, by a finer argument, that if every $n \leq 1536 \cdot 2^{60} = 3 \cdot 2^{69}$ reaches 1, then every nontrivial cycle contains more than $1.375 \cdot 10^{11}$ odd numbers. Barina’s verification passed that range on 3 November 2023 ([3], Table 10), and his 2025 paper (§6) states that at 2^{71} the length of a nontrivial cycle rises to $355,504,839,929$. That number is $m + A$ for $m = 137,528,045,312$ and $A = 217,976,794,617$, the denominator and numerator of the next best upper approximation of $\log_2 3$ after the fraction above (computed here); so the length is counted in steps of $n \mapsto n/2$, $n \mapsto 3n + 1$. With $m > 1.375 \cdot 10^{11}$, the argument of Corollary 2.9 gives $A > m \log_2 3 > 2.179 \cdot 10^{11}$ halvings and $k \geq m(2 - \log_2(3 + 2^{-71})) > 5.706 \cdot 10^{10}$ steps with exponent 1 (Conditional on Hercher’s theorem, whose proof was not read, and on Barina’s computation).

2.5 Restatements, and the Tao-clock deduction

The workbench states, with a proof in Chapter 9 (located; the statement re-derived from the overview in 44a_§2.6), that $\ker(H^1(K_\epsilon) \rightarrow H^1(K))$ for the orbit-graph complex K and its first-jet thickening K_ϵ is $\bigoplus_C \mathbb{Z}/m_C \oplus \bigoplus \mathbb{Z}$, over the nontrivial cycles C of lengths m_C and the components without a cycle, so that the conjecture is equivalent to its vanishing; and (Chapter 3, audited) that it is equivalent to $H_0 = 0$ for a relative orbit-graph homology. These are exact restatements, and the workbench says that the vanishing is not established.

Its Tao-clock preprint (status: conditional on Tao’s Proposition 1.11 [20], which it cites; the deduction audited, the preprint’s repairs to Tao’s §§5–6 not audited) deduces from Tao’s theorem that the laws of the first entry of an orbit into $[1, x]$, started from the logarithmic law on $[t_{j+1}, t_{j+1}^\alpha]$ with $t_j = b^{\alpha^j}$, $\alpha = 1001/1000$, converge in ℓ^1 as $j \rightarrow \infty$, at the rate $\|\lambda_{x,j} - \lambda_x^\infty\| \leq L_c (\log t_j)^{-c}$ for $x \leq t_j$ with $L_c = B'/(1 - \alpha^{-c})$, where $c > 0$ and B' are the constants of Tao’s Proposition 1.11; coherently in x ; and with joint laws for several thresholds converging with the same error, independent of their number (44_ Theorem 44.11). The steps (nested first passage, ℓ^1 contraction, a geometric sum of Tao’s adjacent-scale errors, completeness) were checked; the preprint proves no stronger orbit-minimum bound, as it says.

2.6 The Hurwitz zero-cluster encoding

The workbench uses the Hurwitz zeta function in three places: its split-zero history note, its Chapter 1 §5 (the same reconstruction applied to the history law) and its Chapter 2 §10 (Hurwitz values at nonpositive integers). The first treats the motion of a zero under $\zeta(s, 1 + t)$ in full.

Fix (m, A) ; the K odd-return words, that is the exponent words $w = (a_1, \dots, a_m)$ with all $a_i \geq 1$ and $a_1 + \dots + a_m = A$, have residues r_w (the classes of Proposition 2.1, read modulo 2^{A+1}) that are pairwise distinct. For a probability vector λ on these words put $Z_\lambda(s, t) = \sum_w \lambda_w \zeta(s, 1 + tr_w)$ for $|t|(2^{A+1} - 1) < 1$.

Theorem 2.10. *Let ρ be a nontrivial zero of ζ . The t -jet of order J of the zeros of $Z_\lambda(\cdot, t)$ near ρ (of their monic cluster polynomial, if ρ is multiple) determines the moments $M_i = \sum_w \lambda_w r_w^i$, $i \leq J$, and conversely. Hence the jet of order $K - 1$ determines λ , and for $K \geq 2$ the jet of order $K - 2$ determines λ at no interior point of the simplex: its fibre through every such point contains a segment.*

Status. Proved here for simple zeros; audited in general. The reconstruction is the zeta programme's (historical_hurwitz_jets.tex, equations H10–H13), which the Collatz note cites and re-proves in the part it uses; checked numerically at $\rho = \frac{1}{2} + 14.1347 \dots i$ (the weights (0.5, 0.2, 0.3) at $(m, A) = (2, 4)$ are recovered).

Proof for a simple zero. By the zeta reader's jet theorem (Theorem “the first Hurwitz jet”, (a); [9, 25.11.10]), $\zeta(s, 1+u) = \sum_j b_j(s) u^j$ with $b_j(s) = (-1)^j (s)_j \zeta(s+j)/j!$, so $Z_\lambda(s, t) = \sum_j b_j(s) M_j t^j$ with $M_0 = 1$. For $j \geq 1$, $b_j(\rho) \neq 0$, since ρ is not a nonpositive integer and $\operatorname{Re}(\rho + j) > 1$. Writing the zero as $\rho(t) = \rho + c_1 t + c_2 t^2 + \dots$, the coefficient of t^j in $Z_\lambda(\rho(t), t) = 0$ is $\zeta'(\rho) c_j + b_j(\rho) M_j$ plus a polynomial in $c_1, \dots, c_{j-1}$ and $M_1, \dots, M_{j-1}$ whose coefficients are derivatives at ρ of ζ and of the b_j ; so $(c_1, \dots, c_j)$ and $(M_1, \dots, M_j)$ determine each other triangularly. The K distinct nodes r_w make $M_0, \dots, M_{K-1}$ determine λ (Vandermonde). With $h_w = 1 / \prod_{v \neq w} (r_w - r_v)$, which is nonzero for every w , $\sum_w h_w r_w^i = 0$ for $i \leq K - 2$ (in particular $\sum_w h_w = 0$); so for every λ with all $\lambda_w > 0$ and all small ε , $\lambda + \varepsilon h$ is a probability vector with the moments of λ up to order $K - 2$. $\square$

This is an exact, invertible encoding of a probability vector into the motion of a zeta zero. The family is built from the weights and any nontrivial zero works, so it carries no information about Collatz orbits beyond the residues, and none about the location of the zeros; the note says so. The same note gives three information-loss witnesses: two parity words of length 5 whose affine constant terms $\frac{1}{2} + \frac{q}{32}$ and $\frac{1}{8} + \frac{q}{16} + \frac{q^2}{32}$ differ by $(q-3)(q+4)/32$ and agree at $q = 3$; the eigenvalues of the affine matrix forget bit order; and $(K-2)$ -jets do not separate weights.

2.7 Negative results, with exact scope

- The full-history obstruction (Theorem 2.2(c)): it bounds no coarser observable and says nothing about orbits.
- The encoding losses of §2.6: they do not show that no faithful spectral encoding exists; the order- $(K-1)$ encoding is faithful.
- The formal inverse $(I - tP)^{-1}$ of Chapter 3 is not a finite integral chain on cycles or divergent rays, and the signed primitive of the unit cochain is unbounded below on an acyclic component: these name inputs a global proof would need, not obstructions to the conjecture.
- The finite template search of Chapter 20 does not show that a template applies at every retained source.

3 The Erdős Problem 817 workbench

For a finite set A of distinct positive integers let $H(A) = \{\sum_{a \in A} \varepsilon_a a : \varepsilon_a \in \{0, 1\}\}$ (the empty sum included) and $S(A) = \sum_{a \in A} a$. A is k -admissible if $H(A)$ contains no k -term arithmetic progression with nonzero step, and $g_k(n)$ is the least N such that some n -element $A \subseteq \{1, \dots, N\}$ is k -admissible. Erdős Problem 817, a problem of Erdős and Sárközy, asks for estimates of $g_k(n)$, in particular whether $g_3(n) \gg 3^n$; Erdős and Sárközy proved $g_3(n) \gg 3^n/n^{O(1)}$ [6]. The problem was listed as open on erdosproblems.com/817 when read on 26 September 2026. The workbench reports that Costa's preprint [4] proves $\liminf g_3(n)/3^n = 0$, a negative answer to that question; this reader did not check it, and the workbench's own results concern $k \geq 4$. Admissibility is hereditary, since $H(B) \subseteq H(A)$ for $B \subseteq A$; so g_k is strictly increasing in n .

Credit, as the workbench records it. The $k = 4$ proof is due to a contributor on the subreddit r/LLMmathematics whose account was later deleted. The Clankers reconstructed the argument, made its compressed steps explicit, checked it and formalised its central algebraic and base-19 components in Lean; sneed-and-feed formalised the finite upper bound in Lean (pull request #1). The workbench makes no novelty or priority claim for the $k = 4$ theorem, which it describes as a result about the $k = 4$ exponential-rate subproblem, not a solution of every part of Problem 817. Audit note: 45_ ; first-pass report 45a_.

3.1 The rate $19^{1/3}$ for four-term progressions

Theorem 3.1. *For every $n \geq 1$, $\frac{19^{n/3} - 1}{2n} \leq g_4(n) \leq 8 \cdot 19^{\lfloor n/3 \rfloor - 1}$. Hence $\lim_n g_4(n)^{1/n} = 19^{1/3} \approx 2.6684$.*

Status. Proved here; the workbench's paper ep817_k4_rate.tex, audited: complete and correct as written. The workbench reports the upper half as Lean-checked (Extended.lean); the Lean files were read, not built. The lower bound is not in Lean; only the algebra of Lemma 3.2 is (Core.lean). Lemmas 3.2–3.4 and Corollary 3.6 have the same status.

Let $A = \{a_1, \dots, a_n\}$ be 4-admissible. A relation is $c \in \{-2, \dots, 2\}^n$, $c \neq 0$, with $\sum c_i a_i = 0$; a ± 1 -relation has $c \in \{-1, 0, 1\}^n$.

Lemma 3.2 (relation splitting). *If c is a relation, then with $P_j = \sum_{c_i=j} a_i$ and $N_j = \sum_{c_i=-j} a_i$ one has $P_1 = N_1$ and $P_2 = N_2$: the ± 1 part and the ± 2 part of c are each a relation or zero.*

Proof. $X_0 = P_1 + N_2$, $X_1 = P_1 + P_2$, $X_2 = N_1 + N_2$, $X_3 = N_1 + P_2$ are subset sums, since the four index classes are disjoint. With $d = P_2 - N_2$ the relation $P_1 + 2P_2 = N_1 + 2N_2$ gives $N_1 - P_1 = 2d$, and then $X_1 - X_0 = X_2 - X_1 = X_3 - X_2 = d$. Admissibility forces $d = 0$, and then $P_1 = N_1$. $\square$

Lemma 3.3 (blocks and the short kernel). *Call a ± 1 -relation minimal if its support is inclusion-minimal. Two minimal relations have disjoint supports or are equal up to sign. Every relation is $\sum_j t_j r^{(j)}$ with $t_j \in \{-2, \dots, 2\}$, where the $r^{(j)}$ are the minimal relations (the blocks), which have pairwise disjoint supports of size at least 3.*

Proof. For minimal r, s , the vectors $r + s$ and $r - s$ have entries in $\{-2, \dots, 2\}$ and their ± 2 layers are $r\mathbf{1}_{E_+}$ and $r\mathbf{1}_{E_-}$, where $E_{\pm} = \{i : r_i = \pm s_i \neq 0\}$. By Lemma 3.2 each is a relation or zero. If the supports meet, one of $E_{\pm}$ is nonempty, minimality of r gives $E_{\pm} = \text{supp } r$, so $s = \pm r$ on $\text{supp } r$, and minimality of s gives equality. Every ± 1 -relation u is a sum of blocks with disjoint supports: a relation r with inclusion-minimal support inside $\text{supp } u$ is minimal, the argument just given applied to $u \pm r$ shows $u = \pm r$ on $\text{supp } r$, and $u \mp r$ is then a ± 1 -relation or zero that vanishes on $\text{supp } r$; induction on the support finishes. For a relation c , the ± 1 part u and half the ± 2 part v are ± 1 -relations with disjoint supports (Lemma 3.2), each a sum of whole blocks, so $c = u + 2v = \sum (\alpha_j + 2\beta_j) r^{(j)}$ with $\alpha_j \beta_j = 0$. A block of size 1 would be a zero element, one of size 2 two equal elements. $\square$

Lemma 3.4 (the ternary count). *Let $T(A) = \{\sum x_i a_i : x \in \{0, 1, 2\}^n\}$, let the blocks have sizes $m_1, \dots, m_h$ and let $t = n - \sum m_j$. Then $|T(A)| = 3^t \prod_j (3^{m_j} - 2^{m_j})$.*

Proof. Two vectors in $\{0, 1, 2\}^n$ have the same value iff their difference is a relation, iff it is $\sum t_j r^{(j)}$ (Lemma 3.3). So the fibres are products over the blocks, and the coordinates outside the blocks are fixed. On a block, reflecting $x_i \mapsto 2 - x_i$ where $r_i = -1$ turns the equivalence into translation by $(1, \dots, 1)$, and each translation class in $\{0, 1, 2\}^m$ has exactly one representative with minimum 0; these number $3^m - 2^m$. $\square$

Proof of Theorem 3.1. Lower bound. $f_m = 3^m - 2^m$ has $f_3 = 19$ and $f_{m+1} = 3f_m + 2^m > 3f_m \geq 19^{1/3}f_m$, so $f_m \geq 19^{m/3}$ for $m \geq 3$; also $3 \geq 19^{1/3}$. By Lemma 3.4 $|T(A)| \geq 19^{n/3}$, and $T(A) \subseteq [0, 2S(A)] \subseteq [0, 2nN]$. *Upper bound.* $D = H(\{1, 7, 8\}) = \{0, 1, 7, 8, 9, 15, 16\}$ contains no nonconstant 4-term progression modulo 19 (all $19 \cdot 18$ progressions checked). For $A_m = \{19^j w : j < m, w \in \{1, 7, 8\}\}$, $H(A_m)$ is the set of integers whose base-19 digits lie in D , without carries since $16 < 19$. If $x + id$ ($0 \leq i \leq 3$) lie in $H(A_m)$ with $d = 19^r e$, $19 \nmid e$, then $r < m$, the lowest r digits agree, and the r -th digits $\delta_i \in D$ satisfy $\delta_i \equiv \delta_0 + ie \pmod{19}$, a nonconstant progression in D modulo 19. So A_m is 4-admissible with $3m$ elements and maximum $8 \cdot 19^{m-1}$; for general n take $m = \lceil n/3 \rceil$ and delete elements. $\square$

The top level of this construction needs only to be admissible, not to satisfy a modular condition.

Proposition 3.5 (lifting). *Let each W_j be $\{1, 7, 8\}$ or $\{2, 3, 5\}$. For every 4-admissible set B and every $q \geq 0$, the set $\{19^j w : j < q, w \in W_j\} \cup 19^q \cdot B$ is 4-admissible. Hence:*

- (a) $g_4(n + 3q) \leq 19^q g_4(n)$ for all $n \geq 1, q \geq 0$;
- (b) $g_4(n) \leq c \cdot 19^{\lceil n/3 \rceil - 1}$ with $c = 1, 3, 5$ for $n \equiv 1, 2, 0 \pmod{3}$ (top levels $\{1\}, \{2, 3\}, \{2, 3, 5\}$);
- (c) with Proposition 3.11: $g_4(3q) \leq 79 \cdot 19^{q-2}$ and $g_4(3q+1) \leq 246 \cdot 19^{q-2}$ for $q \geq 2$, and $g_4(3q+2) \leq 40 \cdot 19^{q-1}$ for $q \geq 1$; that is, at most 0.219, 0.256 and 0.296 times $19^{n/3}$, against 0.421, 3.00 and 1.12 times $19^{n/3}$ in Theorem 3.1;
- (d) for every certificate (q, B) of Theorem 3.8, $g_k(m|B| + n) \leq q^m g_k(n)$; for example $g_6(10m + n) \leq 1651^m g_6(n)$, which sharpens Theorem 3.9(b) when $10 \nmid n$.

The sets $\{1, 7, 8\}$ and $\{2, 3, 5\} = 3 \cdot \{1, 7, 8\} \pmod{19}$ are the only 3-element sets B with $S(B) < 19$ and $H(B)$ free of nonconstant 4-term progressions modulo 19.

Status. Generalised here; (c) uses only the upper bounds $g_4(5) \leq 40, g_4(6) \leq 79, g_4(7) \leq 246$, given by the admissible sets $\{1, 13, 35, 39, 40\}, \{2, 29, 45, 74, 77, 79\}$ and $\{1, 3, 39, 180, 219, 243, 246\}$, not the exhaustive searches. Checked: the lifted sets for $q = 1, 2$, both digit sets and seven admissible top levels (28 sets); all 3-element digit sets modulo 19; for $k = 6, A^\# \cup 1651 \cdot B$ for four 6-admissible B , with a non-admissible B as a negative control.

Proof. H of the lifted set is $H(A) + 19^q \cdot H(B)$, where $H(A)$ is the set of integers below 19^q whose j -th base-19 digit lies in $D_j = H(W_j)$, without carries since $S(W_j) \leq 16$. Let $x_i = y_i + 19^q z_i$ ($0 \leq i \leq 3$) be a progression in it with step $d \neq 0$. If $19^q \mid d$, then $y_i \equiv y_0 \pmod{19^q}$ with $0 \leq y_i < 19^q$, so $y_i = y_0$ and (z_i) is a progression with nonzero step in $H(B)$, which is impossible. Otherwise $d = 19^r e$ with $r < q$ and $19 \nmid e$, and, as in the proof of Theorem 3.1, the r -th digits give a nonconstant progression in D_r modulo 19; $H(\{2, 3, 5\}) = 3 \cdot D \pmod{19}$ with 3 a unit, so neither digit set contains one. (a) follows with an optimal B , (b) with the admissible top levels named (their sums of subsets are $\{0, 1\}, \{0, 2, 3, 5\}$ and $\{0, 2, 3, 5, 7, 8, 10\}$), and (c) with the three sets named in the status line. (d) is the same argument with q in place of 19: the r -th digits then form a k -term progression with nonzero step modulo q in $H(B)$, which a certificate excludes. The last statement is a finite search. $\square$

Corollary 3.6 (The Clankers). $g_4(n) \geq \lceil (19^{n/3} + n(n-1) - 1)/(2n) \rceil$, since distinct elements give $S(A) \leq nN - n(n-1)/2$.

3.2 The variance refinement

Theorem 3.7 (EP-03). Write $n = 3q + r$ ($0 \leq r \leq 2$) and $M_n = 19^q 3^r$. Every 4-admissible A has $|T(A)| \geq M_n$ and $19(|T(A)|^2 - 1) \leq 192 \sum a_i^2$. Hence $g_4(n) \geq \lceil \sqrt{19(M_n^2 - 1)/(192n)} \rceil$, about $c_r 19^{n/3} / \sqrt{n}$ with $c_r = 0.315, 0.354, 0.398$ for $n \equiv 0, 1, 2 \pmod{3}$; and, as $T(A) \subseteq [0, 2S(A)]$ with $S(A) \leq nN - n(n-1)/2$, $g_4(n) \geq \lceil (M_n + n(n-1) - 1)/(2n) \rceil$.

Status. Proved here (the variance computation derived independently for this reader; it agrees with the workbench's); the workbench's note, audited; labelled there as not Lean-checked. It rests on the anonymous contributor's Lemmas 3.3–3.4. The last bound is the workbench's (its §6, audited); it improves Corollary 3.6 by replacing $19^{n/3}$ with M_n , a factor $(3/19^{1/3})^{n \bmod 3} \leq 1.264$, and the variance bound is stronger for every $n \geq 3$.

Proof. $|T| \geq M_n$, $f_3 = 19, f_4 = 65 \geq 57, f_5 = 211 \geq 171$ and $f_{m+3} - 19f_m = 8 \cdot 3^m + 11 \cdot 2^m > 0$ give $f_m \geq 19^{\lfloor m/3 \rfloor} 3^{m \bmod 3}$; the product in Lemma 3.4 is then at least $19^\alpha 3^\beta$ with $3\alpha + \beta = n$, and since $3^3 > 19$ the least such value is M_n . *Variance.* By Lemma 3.4 the uniform law on $T(A)$ is the law of a sum of independent components: a free coordinate contributes uniformly from $\{0, a_i, 2a_i\}$ (variance $\frac{2}{3}a_i^2$), a block uniformly from its class values. After the reflection, a block with coefficients $b_i = \pm a_i$, $\sum b_i = 0$, has class values $\langle b, y \rangle$ for $y \in R = \{y \in \{0, 1, 2\}^m : \min y = 0\}$. The map $y \mapsto 2 - y$ permutes classes and negates values, so the mean is 0. Since $\{0, 1, 2\}^m = R \sqcup (1 + \{0, 1\}^m)$ and values are translation invariant, $\sum_{y \in R} \langle b, y \rangle^2 = 3^m \cdot \frac{2}{3}Q - 2^m \cdot \frac{Q}{4}$ with $Q = \sum b_i^2$ (the second moments of $\sum b_i Y_i$ for Y_i uniform on $\{0, 1, 2\}$ and on $\{0, 1\}$). Dividing by $3^m - 2^m$ gives the variance $\kappa_m Q$, $\kappa_m = \frac{8 \cdot 3^m - 3 \cdot 2^m}{12(3^m - 2^m)}$, and $\frac{16}{19} - \kappa_m = \frac{5(8 \cdot 3^m - 27 \cdot 2^m)}{228(3^m - 2^m)} \geq 0$ for $m \geq 3$, with $\kappa_3 = \frac{16}{19} > \frac{2}{3}$. So the variance is at most $\frac{16}{19} \sum a_i^2 \leq \frac{16}{19} n N^2$, while M distinct integers have uniform variance at least $(M^2 - 1)/12$. $\square$

3.3 General k

Theorem 3.8 (EP-04). *For every $k \geq 3$, $\Lambda_k = \lim_n g_k(n)^{1/n}$ exists and equals $\inf q^{1/|B|}$ over the pairs (q, B) with $S(B) < q$ and $H(B)$ modulo q free of k -term progressions with nonzero step (q need not be prime).*

Status. Proved here; the workbench's general- k note, audited, together with its bound $\Lambda_k \geq (k-1)/(k-2)$, the monotonicity $\Lambda_{k+1} \leq \Lambda_k$ and its Theorem 2: whether base- q numbers with digits from a finite set avoid k -term progressions at every length is decided by a finite carry graph. For $k = 3$: the certificate $(3, \{1\})$ (the powers of 3, whose sums of subsets are the integers with ternary digits 0 and 1) gives $\Lambda_3 \leq 3$. Conversely, a 3-admissible A has no relation with coefficients in $[-2, 2]$: in the notation of Lemma 3.2, X_0, X_1, X_2 is a 3-term progression with step $P_2 - N_2$, so $P_2 = N_2$ and $P_1 = N_1$, and a nonzero ± 1 or ± 2 layer would give the progression $0, P_j, 2P_j$. So the 3^n ternary sums are distinct, $3^n \leq 2S(A) + 1 \leq 2nN + 1$, and $\Lambda_3 \geq 3$. Hence $\Lambda_3 = 3 > \Lambda_4 = 19^{1/3} > 97^{1/6} \geq \Lambda_5$: the sequence (Λ_k) decreases strictly at its first two steps (Proved here; the workbench's located chapters 22 and 24 state the injectivity and $\Lambda_3 = 3$ in their notation, and the bound $g_3(n) \geq (3^n - 1)/(2n)$ is of the form of Erdős and Sárközy's $3^n/n^{O(1)}$).

Proof. If A and B are k -admissible and $R = 2S(A) + 1$, then $C = A \cup R \cdot B$ is k -admissible and $2S(C) + 1 = (2S(A) + 1)(2S(B) + 1)$: $H(C) = H(A) + R \cdot H(B)$ bijectively, and a progression $x_i = \alpha_i + R\beta_i$ in $H(C)$ has second differences $\Delta^2 \alpha + R\Delta^2 \beta = 0$ with $|\Delta^2 \alpha| \leq 2S(A) < R$, so α and β are progressions and one is nonconstant. Hence $F_k(n) = \min(2S(A) + 1)$ over k -admissible n -sets is submultiplicative, $\lim F_k(n)^{1/n} = \inf F_k(n)^{1/n}$ by Fekete's lemma, and $2g_k(n) + 1 \leq F_k(n) \leq 2ng_k(n) + 1$. For a certificate (q, B) the sets $\{q^j b : b \in B, j < m\}$ are k -admissible, by the argument of the upper bound in Theorem 3.1 with q in place of 19 ($q \nmid e$ suffices); so $\Lambda_k \leq q^{1/|B|}$. Conversely every k -admissible B gives the certificate $(2S(B) + 1, B)$, because elements of $[0, S(B)]$ whose consecutive differences agree modulo $2S(B) + 1$ have equal integer differences. $\square$

Theorem 3.9 (certificates for $k = 5, 6$). (a) $B^* = \{1, 4, 5, 17, 21, 22\}$ has $S(B^*) = 70$, $|H(B^*)| = 38$, and $H(B^*)$ is free of nonzero-step 5-term progressions modulo 97 and of 6-term progressions modulo 93. Hence $\Lambda_5 \leq 97^{1/6} \approx 2.1435$ and $\Lambda_6 \leq 93^{1/6} \approx 2.1285$ (EP-04, Theorem 3).

(b) $A^\# = \{3, 4, 7, 34, 37, 41, 216, 250, 253, 257\}$, the ten pairwise distances of the marks 0, 4, 7, 41, 257, has $S(A^\#) = 1102 < 1651 = 13 \cdot 127$, $|H(A^\#)| = 291$, and $H(A^\#)$ is free of nonzero-step 6-term

progressions modulo 1651. Hence $g_6(n) \leq 257 \cdot 1651^{\lceil n/10 \rceil - 1}$ and $\Lambda_6 \leq 1651^{1/10} \approx 2.0979$ (EP-05, labelled by the workbench as awaiting independent review).

Status. Audited (certificate): each modular condition recomputed in the first pass and again for this reader; the lifting is Theorem 3.8.

Proposition 3.10 (distance rulers; EP-05). *If a set A of positive integers contains r pairwise disjoint subsets each summing to L , then $H(A)$ contains the $(r + 1)$ -term progression $0, L, \dots, rL$. In particular, for marks $0 = t_0 < \dots < t_{m-1} = L$ whose $2m - 3$ distances $L, t_i, L - t_i$ ($1 \leq i \leq m - 2$) are pairwise distinct (for example, when all $\binom{m}{2}$ pairwise distances are distinct), the subset sums of any set containing these distances contain $0, L, \dots, (m - 1)L$.*

Proof. Sums of j of the disjoint subsets give jL . For the marks, the $m - 1$ sets $\{L\}$ and $\{t_i, L - t_i\}$ are pairwise disjoint and each sums to L . $\square$

Status. The first statement is the case of 0/1 vectors of the workbench's EP-05 Theorem 3.1, which allows signed vectors with pairwise disjoint supports and a common nonzero observed boundary (audited); the ruler statement is Generalised here from its corollary, which assumes all $\binom{m}{2}$ distances distinct. For $A^\#$ it gives $0, 257, 514, 771, 1028$, so $A^\#$ is 6-admissible and not 5-admissible.

Comparison with the literature, as far as read. Korsky's preprint [13] (read in its arXiv HTML version, Theorems 1.2–1.3 and (1.8)) proves, for fixed $k \geq 4$, $g_k(n) \gg_k ((k - 1)/(k - 2))^n n^{-\log_2((k - 1)/(k - 2))}$, and upper bounds whose exponential rate is $\min_p p^{2/(\min\{p, k\} - 1)}$ over primes $p \geq 3$: $5^{2/3} \approx 2.924$ for $k = 4$, $\sqrt{5} \approx 2.236$ for $k = 5$, $7^{2/5} \approx 2.178$ for $k = 6$ (this reader's arithmetic). Against these, Theorem 3.1 determines the $k = 4$ rate, which lies between Korsky's $3/2$ and $5^{2/3}$, and Theorem 3.9 lowers the upper rates for $k = 5, 6$; the workbench presents the latter as improvements on Korsky's rates and disclaims priority for the former. No wider search was made.

3.4 Exact small values

Proposition 3.11 (computed for this audit). $g_4(1), \dots, g_4(6) = 1, 3, 5, 14, 40, 79$, and $176 \leq g_4(7) \leq 246$.

Status. Computed for $n \leq 6$ by three independent exhaustive searches (two in the first pass, one by the author of this reader using heredity pruning only and no lemma of the paper) and a fourth by the referee of this reader; they agree, including the numbers of optimal sets $1, 2, 2, 2, 7, 1$. The unique optimal 6-set is $\{2, 29, 45, 74, 77, 79\}$: two blocks ($29 + 45 = 74$, $2 + 77 = 79$), and $|T| = 361 = M_6$, the least value Theorem 3.7 allows. $g_4(7) \leq 246$ by the admissible set $\{1, 3, 39, 180, 219, 243, 246\}$; the lower bound 176 is conditional on the first-pass search program, which excluded maxima in $[132, 175]$ by pruning derived from Lemma 3.4 and Theorem 3.7; maxima below 132 are excluded by the refined form of Theorem 3.7, with $\sum a_i^2 \leq \sum_{j < 7} (N - j)^2$. The values are not in the workbench or in the sources read for this reader; OEIS could not be consulted. No claim of novelty.

3.5 The continuation notes

The workbench's 29 continuation notes are described as ordinary proofs with exact finite certificates, submitted for independent review, without new Lean. Beyond Theorems 3.8–3.9 and Proposition 3.10:

- *Finite-period approximation* (EP-08; audited in part): for a finite dictionary D of levels, the best lower-limit growth rate $\delta_q(D)$ of the q -ary images satisfies $0 \leq \log U_m - \log \delta_q(D) \leq \log(q - 1)/(m n_{\min})$, where $U_m = \min_{w \in D^m} F_q(w)^{1/N(w)}$ is the least per-generator image rate over words of m levels ($F_q(w)$ the image size and $N(w)$ the number of generators of the word) and $n_{\min}$

the least block size; the radix-ten example has minima $a_1 = 13$, $a_{3r} = 893^r$, $a_{3r+1} = 93^2 \cdot 893^{r-1}$ ($r \geq 1$), $a_{3r+2} = 93 \cdot 893^r$ (verified for $m \leq 16$ by an independently built automaton). This is an image-count rate, and the workbench says $893^{1/6}$ is not a record.

- *A published correction* (audited): in the seven-observable note the independence test $Y_P = P \cup (L - P)$ with $P \subseteq [0, Q]$ needs $L > 3Q$, not the older $L > 2Q$.
- *Recurrence control R1* (audited): if $a_j > S_{j-1} + S_{j-r}$ for all j , the subset sums contain no $(2^r + 1)$ -term progression; for $r = 2$ this covers the Pell numbers 1, 2, 5, 12, 29, ...
- The outer control, the cone dual (except its statement that image counts do not decide admissibility, audited), the rest of the recurrence notes and the all-block image theorem are located.

3.6 Negative results, with exact scope

1. The “19 classes per 3-block” count needs admissibility: $\{1, 2, 3\}$ has $|T| = 13 < 19$.
2. Admissible classes are strictly nested: $\{1, 2\}$ is 5-admissible but not 4-admissible.
3. Modular certificates are sufficient, not necessary, for all-length admissibility ($B = \{2, 7\}$, $q = 11$, $k = 3$); Theorem 3.8 is unaffected.
4. For B^* every base 71–96 fails at $k = 5$ and every base 71–92 at $k = 6$ (the two-level language already contains a 5-term, respectively 6-term, progression): a statement about this block only.
5. Every distinct-distance ruler with m marks has an m -term progression (Proposition 3.10); the ruler $(0, 1, 5, 22)$ has no 6-admissible extension by one further mark with distinct distances.
6. For $A^\#$ no canonical schedule with radix in $[1103, 1650]$ is admissible; smaller radices and other blocks are outside this statement.
7. Image counts do not determine admissibility.

None of these bears on the unrestricted Λ_k for $k \geq 5$, which the workbench leaves open.

4 The Yang–Mills workbench, with the Navier–Stokes, S^6 and Jacobian-map material

The workbench yang-mills-interacting-workbench is an open, AI-assisted workbench of the owner’s, with work done with ChatGPT 5.6 Sol and GPT-6 Astra (README); the source results it imports keep their own attribution (ATTRIBUTION.md). It does not claim a solution of the interacting four-dimensional problem, says that the continuum mass-gap question remains unresolved in the collection, and claims no Lean or independent external analytical certification. Audit notes: 46_ (and 21_, 28_, 34_); first-pass report 46a_.

4.1 The Hamiltonian and the gap

Box and space. The vertices form the open box $\{-L, \dots, L\}^3$, $L \geq 2$; the links e are the positively oriented nearest-neighbour edges; the plaquettes p are the elementary faces, $M_L = 12L^2(2L + 1)$ of them. The configuration space carries one $SU(2)$ matrix U_e per link, with the product Haar probability measure; the *physical* space is the subspace of L^2 invariant under the gauge transformations at every vertex, boundary vertices included.

Hamiltonian (the workbench’s normalisation). $H_L = \kappa K + \kappa \xi (2M_L - S)$ with $\kappa = 2g^2/a$, $\xi = 1/(4g^4)$, $a > 0$ the spacing and $g > 0$ the bare coupling. $K = -\sum_{e,a} X_{e,a}^2$ is the sum of the link Casimirs; on the Peter–Weyl block with link spins $j = (j_e)$ it acts as $c_j = \sum_e j_e(j_e + 1)$. $S = \sum_p W_p$ with W_p the trace of the ordered product around p , so $\kappa \xi (2M_L - S) = (2g^2a)^{-1} \sum_p (2 - W_p) \geq 0$. This is the

Kogut–Susskind Hamiltonian [12]; the workbench records the conversion to the convention of Schütte, Zheng and Hamer [18], so thresholds such as $g^2 \geq 3.68$ below are in the workbench’s normalisation.

Gap. Δ_L is the distance from the ground energy E_0 to the rest of the spectrum of H_L on the physical space. At $\xi = 0$ it equals 3κ : one plaquette excited in the fundamental representation, four links of Casimir $3/4$.

Theorem 4.1 (YM-01). *Let (m_i, t_i) , $i = 1, \dots, 5$, be the ten rational numbers of the workbench’s F26, $\ell_5(x) = \sum (m_i + 4t_i)x^i$, $\delta_5(x) = \sum_{i,j \leq 5, i+j \geq 6} 3(m_it_j + m_jt_i)x^{i+j}$, $\mathcal{D}_5 = (1 - \ell_5)^2 - \frac{8}{3}\delta_5$, α_5 the first positive root of $\mathcal{D}_5$, and*

$$d_5(x) = \frac{3}{2}(1 + \sqrt{\mathcal{D}_5(x)}) + \sum_{i=1}^5 (\frac{3}{2}m_i - 6t_i)x^i.$$

If the (m_i, t_i) bound the local norms of the vacuum coefficients as the workbench’s computations assert, then for every $L \geq 2$, every $a > 0$ and every $0 < \xi \leq \alpha_5$,

$$\Delta_L \geq \kappa d_5(\xi).$$

In the coupling,

$$\xi \leq \alpha_5 \iff g^2 \geq \frac{1}{2\sqrt{\alpha_5}} = 3.6835519839857273 \dots, \quad \alpha_5 = 0.0184249535761176166818 \dots;$$

d_5 decreases on $[0, \alpha_5]$ from $d_5(0) = 3$ (at $\xi = 0$, $\Delta_L = 3\kappa$) to $d_5(\alpha_5) = 1.5453 \dots$; so $\Delta_L \geq 1.5453 \kappa$ for all $g^2 \geq 3.6836$, and, as $d_5(1/64) = 1.9068301567 \dots$, $\Delta_L > 1.9068 \kappa$ for all $g^2 \geq 4$, uniformly in the box and the spacing. Equivalently, $q_{H_L - E_0}(\psi f) \geq \kappa d_5(\xi) \text{Var}_{\psi^2}(f)$ for f in the physical form domain, ψ the normalised ground state (the workbench’s form, F37).

Status. Conditional on the computed coefficient bounds of orders 3, 4 and 5 for the threshold 3.6836 and the constants d_5 ; its qualitative content, a gap on the whole physical space, uniform in the box and the spacing, above an explicit coupling, holds with audited inputs only for $g^2 \geq 4.0187$ (Corollary 4.5). Order 1 is verified by hand; order 2 was recomputed independently in the first pass from explicit $SU(2)$ coefficient tensors (the exact values $m(v_2) = 134.07 \dots$, $t(v_2) = 19.80 \dots$ lie below the workbench’s 5834/39 and 137/6); order 5 (662 representatives, 124,864 anchored multisets, 5,726 rational primal/dual certificates) was reproduced by re-running the workbench’s producer and auditor with byte-identical output, which the workbench had not done at publication, and its logic was read, not re-implemented; orders 3–4 are inherited from earlier continuations and pinned by hash. The analytic argument is audited. α_5 , the threshold and d_5 were recomputed by exact bisection and by root isolation, and the monotonicity of d_5 on a 401-point grid.

4.2 The proof: three lemmas and the analytic core

Lemma 4.2 (gauge-invariance Casimir inequality). *Let the links form a finite graph without loops (parallel links allowed) in which every cycle has length at least g ; the cubic box has $g = 4$. On every Peter–Weyl block j that contains a nonzero gauge-invariant vector, $\sum_f j_f \geq g j_e$ for every link e . Hence $c_j \geq \frac{3g}{2} j_e$ for every e , $\sum_e j_e \leq \frac{2}{3} c_j$, and $c_j \geq \frac{3g}{4}$ on the nonconstant physical space; for the box, $c_j \geq 6j_e$ and $c_j \geq 3$. The first inequality holds for all real weights $j_f \geq 0$ that satisfy the vertex inequalities $j_f \leq \sum_{f' \ni v, f' \neq f} j_{f'}$, and it is an equality for $j = \frac{1}{2}$ on a cycle of length g through e and 0 elsewhere.*

Status. Generalised here: the workbench’s F7 (audited) is the case $g = 4$ on the cubic lattice, proved there from the absence of triangles; the sharp constant on any graph is its girth. Checked exhaustively over half-integer spins with vertex singlets on the cube graph (1,012 nonzero assignments with every $j_e \leq 1$; constant 4), $K_{3,3}$ (4), the 5-cycle and the Petersen graph (5), the 6-cycle (6) and a triangle (3); and by

linear programming over real weights with the vertex inequalities only, on 13 graphs of girth 3 to 8 (among them the Petersen, Heawood, McGee and Tutte–Coxeter graphs, the dodecahedron and the box $\{-1, 0, 1\}^3$), where the optimum is the girth on every link, on three parallel links (2), and on the bridge between two triangles, where it is 4: the constant is the girth of the graph, not the length of the shortest cycle through e .

Proof. A block contains a nonzero invariant exactly when, at every vertex v , the tensor product of the representations of spins j_f ($f \ni v$) contains the trivial representation; by the Clebsch–Gordan rule this gives the vertex inequalities, which are all that is used. *A matching at each vertex.* At v , the inequalities say that no j_f exceeds half of $J_v = \sum_{f \ni v} j_f$. That is the condition for a transport plan between the links at v , with supplies and demands j_f and the diagonal forbidden, by Gale’s supply–demand theorem: a set of two or more columns can be served by every row, and a single column f by the rows $f' \neq f$, which carry $J_v - j_f \geq j_f$. Symmetrising the plan gives a symmetric $M^v \geq 0$ with zero diagonal and row sums j_f . *A circulation.* Give each orientation (dart) of a link f the weight $j_f/2$, and let the dart arriving at v along f pass to the dart leaving v along $f' \neq f$ with weight $M_{ff'}^v/2$. The row and column sums of M^v make this a circulation on the darts, which is therefore a nonnegative combination $\sum_C w_C [C]$ of cyclic dart sequences C , that is, closed walks in which consecutive links (cyclically) differ. Counting weights, $\sum_C w_C |C| = \sum_f j_f$ and $\sum_C w_C n_e(C) = j_e$, where $n_e(C)$ is the number of traversals of e by C . *Each walk.* Cut a walk C with $n_e(C) \geq 1$ at its traversals of $e = uv$. Each piece avoids e and is nonempty, since two traversals of e are never consecutive and $u \neq w$. A piece that ends at the endpoint of e where it started is a closed walk in which consecutive links differ, so it contains a cycle and has length at least g ; a piece joining u and w contains a path, which with e forms a cycle, so it has length at least $g - 1$. Hence $|C| \geq g n_e(C)$, and summing, $\sum_f j_f \geq g j_e$. *The Casimir.* Every nonzero spin is at least $\frac{1}{2}$, so $j_f(j_f + 1) \geq \frac{3}{2}j_f$; hence $c_j \geq \frac{3}{2} \sum_f j_f$, which is $\sum_f j_f \leq \frac{2}{3}c_j$, and $c_j \geq \frac{3g}{2}j_e$. On a nonconstant physical block some $j_e \geq \frac{1}{2}$, so $c_j \geq \frac{3g}{4}$. For the equality case, $j = \frac{1}{2}$ on a cycle of length g satisfies every vertex inequality and has $\sum_f j_f = \frac{g}{2} = g j_e$. $\square$

Lemma 4.3 (ground-state transform). *Let $\Gamma(f, h) = \sum_{e,a} (X_{e,a} f)(X_{e,a} h)$, P_H the Haar mean and $Q_H = I - P_H$. For real v , e^v is an eigenfunction of H_L with eigenvalue E iff $Kv = \zeta S + Q_H \Gamma(v, v)$ and $E = \kappa(2M_L \zeta - P_H \Gamma(v, v))$; then $(H_L - E)(e^v h) = \kappa e^v (Kh - 2\Gamma(v, h))$ for every h .*

Status. Proved here; the workbench’s F33–F36, audited.

Proof. From $X^2(e^v h) = e^v(h(Xv)^2 + 2(Xv)(Xh) + hX^2v + X^2h)$ one gets $K(e^v h) = e^v(h(Kv - \Gamma(v, v)) - 2\Gamma(v, h) + Kh)$. With $h = 1$, $H_L e^v = \kappa e^v(Kv - \Gamma(v, v) + \zeta(2M_L - S))$, which equals $E e^v$ exactly when $Kv - \Gamma(v, v) - \zeta S$ is the constant $E/\kappa - 2M_L \zeta$; taking Haar means ($P_H Kv = 0$, $P_H S = 0$) identifies the constant as $-P_H \Gamma(v, v)$. Inserting this into the formula for $K(e^v h)$ gives the last identity. $\square$

The ground state is unique and positive, so a solution v of the vacuum equation gives the ground state $e^v/\|e^v\|$. With $v_1 = S/3$ (each W_p has Casimir 3) and $B = K^{-1}Q_H \Gamma$ the equation is $v = \zeta v_1 + B(v, v)$, whose formal solution is $\sum \zeta^n v_n$ with $v_n = \sum_{i+j=n} B(v_i, v_j)$; this exponential (coupled-cluster) ansatz is the one the workbench credits, for Hamiltonian lattice field theory, to Schütte, Zheng and Hamer [18].

The analytic core (the workbench’s F4–F32; audited). In the Fourier-algebra norm $\|f\|_X = \sum_j \|A_j(f)\|_1$, which is submultiplicative (Eymard [8]), and in local weighted norms $m(\cdot)$, $t(\cdot)$, $\|\cdot\|_{\text{loc}}$ anchored at a link, $\|B(f, h)\|_{\text{loc}} \leq 3(m(f)t(h) + m(h)t(f)) \leq \frac{2}{3}\|f\|_{\text{loc}}\|h\|_{\text{loc}}$, by Lemma 4.2. With $q_5 = \sum_{i \leq 5} \zeta^i v_i$, the residual $\zeta v_1 + B(q_5, q_5) - q_5$ has exactly the degrees 6–10 and local norm at most $\delta_5(\zeta)$, and $J_5 = 2B(q_5, \cdot)$ has norm at most $\ell_5(\zeta)$. For $0 < \zeta \leq \alpha_5$ a Neumann inverse and a binary-tree majorant give $v = q_5 + w$ with $\|w\|_{\text{loc}} \leq w_* = \frac{3}{4}(1 - \ell_5 - \sqrt{\mathcal{D}_5})$, the small root of

$\frac{2}{3}w^2 - (1 - \ell_5)w + \delta_5 = 0$, endpoint included. The bounds are uniform in the box: each v_n is a sum of universal cluster functions over plaquette multisets, so the anchored sums of the infinite lattice bound every box; the orientation dependence of the Fourier-algebra norm is handled by taking the maximum over the eight axis reflections.

Proposition 4.4 (spectral return). *Under the analytic core, every eigenvalue $\lambda > 0$ of $H_L - E_0$ on the physical space satisfies $\lambda \geq \kappa d_5(\xi)$.*

Status. Proved here, given the analytic core; the workbench's F35–F37, audited.

Proof. Let φ be a physical eigenvector, $h = \varphi/\psi$ (gauge invariant) and $h' = Q_H h \neq 0$ (as $\lambda \neq 0$, h is not constant). By Lemma 4.3 and $\Gamma(v, 1) = 0$, $(K - \lambda/\kappa)h' = 2Q_H \Gamma(v, h')$. If $\lambda/\kappa \geq 3$ there is nothing to prove, since $d_5 \leq 3$. Otherwise $c_j \geq 3$ on every nonconstant physical block (Lemma 4.2), so $c_j - \lambda/\kappa \geq c_j(1 - \lambda/(3\kappa))$ and $\|(K - \lambda/\kappa)h'\|_X \geq (1 - \lambda/(3\kappa))\|Kh'\|_X$. The core gives $\|2\Gamma(v, h')\|_X \leq 4t(v)\|Kh'\|_X \leq \chi\|Kh'\|_X$ with $\chi = 4 \sum t_i \xi^i + \frac{2}{3}w_*$, and Q_H does not increase $\|\cdot\|_X$. Hence $\lambda \geq 3\kappa(1 - \chi)$, and substituting w_* shows $3(1 - \chi) = d_5(\xi)$. $\square$

Nothing in this argument uses the truncation order.

Corollary 4.5 (the gap at every truncation order). *For $1 \leq N \leq 5$ let $\ell_N(x) = \sum_{i \leq N} (m_i + 4t_i)x^i$, $\delta_N(x) = \sum_{i,j \leq N, i+j \geq N+1} 3(m_i t_j + m_j t_i)x^{i+j}$, $\mathcal{D}_N = (1 - \ell_N)^2 - \frac{8}{3}\delta_N$, α_N the first positive root of $\mathcal{D}_N$ and $d_N(x) = \frac{3}{2}(1 + \sqrt{\mathcal{D}_N(x)}) + \sum_{i \leq N} (\frac{3}{2}m_i - 6t_i)x^i$. If (m_i, t_i) , $i \leq N$, bound the local norms of the first N vacuum coefficients, then $\Delta_L \geq \kappa d_N(\xi)$ for every $L \geq 2$, $a > 0$ and $0 < \xi \leq \alpha_N$, and d_N decreases on $[0, \alpha_N]$. In particular:*

(a) ($N = 1$) $\mathcal{D}_1(\xi) = 1 - 256\xi/3$ exactly and $\frac{3}{2}m_1 - 6t_1 = 0$, so

$$\Delta_L \geq \frac{3}{2}\kappa \left(1 + \sqrt{1 - \frac{64}{3g^4}}\right) \quad \text{for all } g^2 \geq 8/\sqrt{3} \approx 4.6188;$$

so $\Delta_L \geq \frac{3}{2}\kappa$ there, with 2.0745κ at $g^2 = 5$ and 2.8304κ at $g^2 = 10$.

(b) ($N = 2$) $\alpha_2 = 0.0154800849822 \dots$, that is $g^2 \geq 4.0186791538833 \dots$, and $\Delta_L \geq \kappa d_2(\xi) \geq \kappa d_2(\alpha_2) = 1.52094\kappa$ on that range (1.7665κ at $g^2 = 4.1$, 2.3033κ at $g^2 = 5$).

(c) ($N = 4$) $\alpha_4 = 0.0181049722316 \dots$, that is $g^2 \geq 3.7159603625 \dots$, with $d_4(1/64) = 1.898118$ and $d_4(4/225) = 1.658436$.

Status. Proved here from the audited core. (a) is the workbench's own order-1 theorem, in its 14–16 September continuation YM-06 (consolidation/20260916, equations G22 and G26–G29, with $\Delta_L > 2.07445\kappa$ at $g^2 \geq 5$), whose own proof was not read; its input, order 1, is verified by hand. (b) is Generalised here relative to the workbench's single-norm order-2 theorem in the same continuation (R10: $\Delta_L \geq \frac{3}{2}\kappa(1 + \sqrt{1 - \frac{256}{3}\xi + \frac{10720}{9}\xi^2})$ for $g^2 \geq \sqrt{(32 + \sqrt{354})/3} \approx 4.1156$): with the audited two-norm pair (m_2, t_2) and no further input, the threshold falls to 4.0187 and the endpoint constant rises from $\frac{3}{2}$ to 1.52094. The same continuation reaches $g^2 \geq 3.8260$ at order 2 (its L17) by a sharper computed bound for the cubic term of the residual (944984/351 in its L10, against the generic $6(m_1 t_2 + m_2 t_1) = 300672/39$), which was not audited. So the qualitative content of Theorem 4.1 holds with audited inputs only for $g^2 \geq 4.0187$, and orders 3–5 lower the threshold to 3.6836 and raise the constants. (c) is conditional on the order-3 and order-4 inputs and does not use order 5. α_N , the thresholds and the values of d_N computed here from the exact rationals of F26 by root isolation; the monotonicity checked on a 401-point grid for $N = 1, \dots, 5$. The workbench's YM-03 edition uses the radius $3/256 = \alpha_1$ for its row remainder.

Proof. Replace q_5 by $q_N = \sum_{i \leq N} \zeta^i v_i$ in the analytic core. By $v_n = \sum_{i+j=n} B(v_i, v_j)$ for $2 \leq n \leq N$, the residual is $\zeta v_1 + B(q_N, q_N) - q_N = \sum_{i,j \leq N, i+j \geq N+1} \zeta^{i+j} B(v_i, v_j)$, of local norm at most $\delta_N(\zeta)$, and $\|2B(q_N, \cdot)\| \leq \ell_N(\zeta)$. On $[0, \alpha_N]$, $\ell_N < 1$: $\ell_N(0) = 0$, and $\ell_N(\zeta_1) = 1$ with $0 < \zeta_1 \leq \alpha_N$ would give $\mathcal{D}_N(\zeta_1) = -\frac{8}{3}\delta_N(\zeta_1) < 0$. So the Neumann inverse and the tree majorant give $\|w\|_{\text{loc}} \leq w_* = \frac{3}{4}(1 - \ell_N - \sqrt{\mathcal{D}_N})$, and Proposition 4.4 with $\chi = 4 \sum_{i \leq N} t_i \zeta^i + \frac{2}{3}w_*$ gives $\lambda \geq 3\kappa(1 - \chi) = \kappa d_N(\zeta)$. Differentiating $\frac{2}{3}w_*^2 - (1 - \ell_N)w_* + \delta_N = 0$ gives $w'_* \sqrt{\mathcal{D}_N} = \delta'_N + \ell'_N w_* > 0$ on $(0, \alpha_N)$, so χ increases and d_N decreases. For $N = 1$, $(m_1, t_1) = (\frac{64}{3}, \frac{16}{3})$ gives $\ell_1 = \frac{128}{3}\zeta$, $\delta_1 = 6m_1 t_1 \zeta^2 = \frac{2048}{3}\zeta^2$ and $\mathcal{D}_1 = 1 - \frac{256}{3}\zeta + (\frac{16384}{9} - \frac{16384}{9})\zeta^2$; with $\zeta = 1/(4g^4)$, $\frac{256}{3}\zeta = \frac{64}{3g^4}$. For $N = 2$, $\frac{3}{2}m_2 - 6t_2 = \frac{3408}{39} > 0$, so $d_2 > \frac{3}{2}$ on the whole range even before the value $d_2(\alpha_2)$ is computed. $\square$

What the theorem is and is not. It is a finite-regulator, strong-coupling statement: fixed spacing, any box, bare coupling above a threshold. The continuum limit needs $a \rightarrow 0$ with $g \rightarrow 0$, the opposite regime, and the workbench records that its standing continuum path eventually leaves the domain. Nothing about the continuum mass gap follows, and the workbench claims nothing. Mass gaps at strong coupling are classical for lattice gauge theories in the Euclidean setting (Osterwalder–Seiler [17]; cited, not re-read); the workbench's statement is an explicit bound, uniform in the box, on the whole physical space of the Kogut–Susskind $SU(2)$ Hamiltonian on open boxes, with a closed coupling threshold; whether a bound of this form is in the literature was not searched. If m_5 and t_5 were both underestimated by a factor 2, the threshold would move to $g^2 \approx 3.737$ and $d_5(1/64)$ to 1.8946; by Corollary 4.5(b), errors of any size in the order-3 to order-5 inputs leave a gap above 1.52κ for $g^2 \geq 4.0187$. The Riemann, Jacobian and S^6 material plays no role in this theorem.

4.3 The other results of the workbench

- YM-05 (17 September; $\Delta_L > 1.6584\kappa$ for $g^2 \geq 15/4$, by the same method with four orders). It is Corollary 4.5(c) at $\zeta = 4/225$: conditional on the order-3 and order-4 inputs, not on order 5. Theorem 4.1 gives $d_5 > 1.6993$ at $g^2 = 15/4$.
- YM-02 (21 September; audited in part). The connected plaquette heat correlations equal their $\zeta^0, \zeta^2, \zeta^4$ terms up to $\frac{67896}{169}e^{-13\tau/8}(55|\zeta|)^6/(1 - (55|\zeta|)^2)$ for $|\zeta| < 1/55$, uniformly in L (the parity step checked by hand: a staggered centre map multiplies every W_p by -1 , so the two-point function is even in ζ); for $g^2 \geq 13$, relaxing the whole complementary physical space changes the plaquette-family energy by less than $1/2000$ (less than $1/25000$ for $g^2 \geq 16$), uniformly in L and a ; a fixed-spacing infinite-volume limit (an outline).
- YM-03 (21 September): the same kind of statements with weaker constants; superseded by YM-02; located.
- YM-06 (14–16 September; located, proofs not read): the gauge-native order-1 gap theorem (G22, G26–G29), which is Corollary 4.5(a); the single-norm order-2 theorem R10 ($g^2 \geq 4.1156$); and the order-2 theorem L17 ($g^2 \geq 3.8260$) with a computed cubic-residual bound.

Proposition 4.6 (a method obstruction; YM-07 §27). *Write $H = (2g^2/a)(H_0 - \zeta W_L) + \text{const}$ on the full space (no gauge projection), with $H_0 = \sum_e E_e$ the sum of the link Casimirs E_e and $W_L = \sum_p W_p$. The operator-norm Neumann series certifies the isolated rank-one ground-state projection on the circle $|z| = 3/8$ exactly for $|\zeta| < 3/(16M_L) = 1/(64L^2(2L + 1))$, and on no contour separating 0 from the rest of the spectrum does the norm-product criterion give more. On the physical space the same route gives exactly $|\zeta| < 3/(4M_L)$, on the circle $|z| = 3/2$.*

Status. Proved here; the full-space statement audited (YM-07 §27); the physical-space radius and the optimality over contours are Generalised here.

Proof. $|W_p| \leq 2$ with equality at $U \equiv I$, so $\|\zeta W_L\| = 2|\zeta|M_L$; the same holds on the physical space, since for every $\varepsilon > 0$ the indicator of the nonempty open gauge-invariant set $\{W_L > 2M_L - \varepsilon\}$ is a physical vector. The spectrum of H_0 is contained in $\{0\} \cup [3/4, \infty)$, so $\|(H_0 - z)^{-1}\| \leq 8/3$ on $|z| = 3/8$, and the series converges when $2|\zeta|M_L \cdot \frac{8}{3} < 1$. Both factors are attained ($\|(H_0 - z)^{-1}\| = 8/3$ at $z = \pm 3/8$), so the norm-product criterion fails beyond this radius. Any contour separating 0 from the rest of the spectrum crosses $(0, \frac{3}{4})$, since the eigenvalue $\frac{3}{4}$ must have winding number 0; at a crossing point x the resolvent norm is $\max\{1/x, 1/(\frac{3}{4} - x)\} \geq \frac{8}{3}$. On the physical space the spectrum of H_0 lies in $\{0\} \cup [3, \infty)$ by Lemma 4.2, with 3 attained by one plaquette in spin $\frac{1}{2}$, and the same two arguments give the resolvent norm $\frac{2}{3}$ on $|z| = \frac{3}{2}$ and the radius $3/(4M_L)$. $\square$

The radius shrinks with the box, so at any fixed coupling this route eventually fails; it does not show that the true analyticity radius is small, and it proves neither a gap nor its absence. It explains why Theorem 4.1 needs local norms: at $L = 2$ ($M_2 = 240$) the global route certifies only $|\zeta| < 1/1280$ on the full space and $|\zeta| < 1/320$ on the physical space, both far inside $\zeta \leq \alpha_5 \approx 1/54.3$. (“Weak-coupling disk” in the source means small $|\zeta|$, that is large g : the regime of Theorem 4.1.)

4.4 The Jacobian map and the low-mode theorem

The map

$$F(x, y, w) = ((1 + xy)^3 w + y^2(1 + xy)(4 + 3xy), y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy), 2x - 3x^2 y - x^3 w)$$

has $\det DF = -2$ and three preimages of $(-\frac{1}{4}, 0, 0)$, namely $(0, 0, -\frac{1}{4})$ and $(\pm 1, \mp \frac{3}{2}, \frac{13}{2})$ (checked symbolically). The workbench’s attribution file identifies it with the counterexample to the Jacobian conjecture announced by Levent Alpöge on 20 July 2026, whose announcement, as the file records, credits Akhil for the question and Fable for the work; it notes that the S^6 construction is a different object. That Keller maps give incompressible polynomial flows, and that the curve $\gamma(\tau) = (z^{-1}, -3z/2, 13z^2/2)$, $z = \sqrt{1 - 8\tau}$, escapes at $\tau = 1/8$, is proved in the zeta reader (Lemma “Keller maps give incompressible polynomial flows”). The workbench’s chain $F \rightarrow$ incompressible flow $\rightarrow$ material tensor $\rightarrow$ lattice local-energy weights $\rightarrow$ trial states ends in its Theorem 14.2, audited in 34_: it is a valid diagonal argument, conditional on fixed-box weak-coupling limits that the file only sketches, and its energy quotients equal, up to a factor $1 \pm 1/(10j)$, the lowest free two-gluon colour-singlet energy of an open box of side $j/50$ at couplings $g_j < 1/j$. It bears on the mass gap in neither direction, and the data of F move the quotients only inside a window that does not depend on them.

4.5 Navier–Stokes

- *The imported theorem* (not audited). The workbench records OpenAI’s manuscript *Finite Time Blowup for Navier–Stokes* [16], Theorem 1.1 (a compactly supported smooth force and zero initial velocity giving a smooth solution on $[0, 1)$ with bounded energy and $\limsup \|u(t)\|_\infty = \infty$, for every viscosity), holds a transcription and a reconstruction reader, and records that complete independent analytic and Lean validation remains unfinished. Every use of it below or elsewhere is conditional on it.
- *Exact constraint data* (NS-04(a); audited in part: the encoding, the decoder and the barrier inequalities checked symbolically; the existence of the Lichnerowicz solution by the standard barrier method read, not re-checked). Every compactly supported smooth divergence-free V on $\mathbb{R}^3$ gives exact vacuum constraint data in four spatial dimensions with $\Lambda = -6/L^2$, through a Lichnerowicz equation with barrier solutions, and an explicit decoder returns V on marked data. It is an encoding of a velocity snapshot, not a dynamical correspondence: inserting the Navier–Stokes time as Einstein time gives a nonzero defect, as the workbench says.

- *Strain and Kasner exponents* (NS-04(b); proved here). For a tracefree symmetric 3×3 matrix B put $\chi = \sqrt{1 + \frac{4}{3} \operatorname{tr} B^2}$, $P_w = \frac{1}{4} - \frac{3}{4\chi}$ and $P_\perp = \frac{I}{4} + (\frac{I}{4} + B)/\chi$. Then $P_w + \operatorname{tr} P_\perp = 1$ and $P_w^2 + \operatorname{tr} P_\perp^2 = 1$, $B = (3P_\perp + (P_w - 1)I)/(1 - 4P_w)$, and $B \mapsto (P_w, P_\perp)$ is a bijection onto the block-Kasner data with $P_w \in [-\frac{1}{2}, \frac{1}{4}]$. *Proof.* $\operatorname{tr} P_\perp = \frac{3}{4} + \frac{3}{4\chi}$ gives the first identity; $\operatorname{tr} P_\perp^2 = \frac{3}{16}(1 + \frac{1}{\chi})^2 + \operatorname{tr} B^2/\chi^2$ gives the second after $\frac{3}{4}(1 + \frac{4}{3} \operatorname{tr} B^2) = \frac{3}{4}\chi^2$; the inverse formula is direct $(3P_\perp + (P_w - 1)I) = 3B/\chi$ and $1 - 4P_w = 3/\chi$; and on the Kasner set $1 + \frac{4}{3} \operatorname{tr} B^2 = 9/(1 - 4P_w)^2$. (In the workbench $B = -2\nu S/c^2$ for a strain S .)
- *A modified equation* (NS-04(d); audited in part). The Navier–Stokes equation with an added positive Fourier multiplier from an elliptic slab extension, whose total damping is at least $c_*|k|^3$ at high frequency, has global smooth solutions: the standard hyperdissipative mechanism (dissipation of order $|k|^{2\alpha}$ with $\alpha \geq 5/4$ [14]), cited by the workbench through Tao [21] without priority. It concerns a modified equation, not Navier–Stokes.

4.6 S^6

The workbench’s `s6/` folder holds a five-page reader of five results of the S^6 project, whose proofs are in the Zenodo archive 10.5281/zenodo.22678442 and not in the repository. The construction is the manuscript [1]: a compact complex threefold fibred by tori over the projective line, which the manuscript states is diffeomorphic to S^6 . That statement was not checked here, and the workbench does not certify a global complex structure on S^6 ; it draws no mass-gap conclusion. The workbench’s `ATTRIBUTION.md`, which it designates as the clarification, says that the manuscript was circulated by Levent Alpöge and produced with Claude (older files say “with Fable”).

Proposition 4.7 (*S6-5, arithmetic part*). *For $a \in \mathbb{Z}^4$ let $P(a) = a_0(a_0^2 - 3(a_1^2 + a_2^2 + a_3^2))$, the real part of a^3 for a read as a quaternion. If $3 \mid P(a)$ then $9 \mid P(a)$; so P never takes a value $3k$ with $3 \nmid k$. On $D_4 = \{a \in \mathbb{Z}^4 : \sum a_i \text{ even}\}$ the values of P are even and generate $2\mathbb{Z}$, so they avoid every $6k$ with $3 \nmid k$ (in particular ± 6). The lattice $\{\sqrt{2}(a, a, a) : a \in D_4\}$ has Gram matrix $6G_{D_4}$, determinant $6^4 \cdot 4 = 5184$ and minimum 12.*

Status. Generalised here: the workbench’s *S6-5* states that $12\sqrt{2}$ is not attained; the argument (the workbench’s own mod-3 step) excludes every $12\sqrt{2}k$ with $3 \nmid k$ and needs no parity for the mod-9 statement. Given the source’s identification of the octonionic cubic on this lattice as $2\sqrt{2}P(a)$ (not checked here), its values generate $4\sqrt{2}\mathbb{Z}$ and never equal $12\sqrt{2}k$ with $3 \nmid k$. Checked for $|a_i| \leq 12$. The literal intersection with Wilson’s Leech lattice and the results *S6-1* to *S6-4* are located.

Proof. Modulo 3, $P \equiv a_0^3$, so $3 \mid P$ forces $3 \mid a_0$, and then $P = 9b(3b^2 - a_1^2 - a_2^2 - a_3^2)$ with $a_0 = 3b$. On D_4 : if a_0 is odd, $a_1 + a_2 + a_3$ is odd, so $a_1^2 + a_2^2 + a_3^2$ is odd and $a_0^2 - 3(\dots)$ is even; if a_0 is even, so is P ; and $P(1, 1, 0, 0) = -2$. An even value divisible by 3 is divisible by 18, so no value is $6k$ with $3 \nmid k$. The Gram matrix of $\sqrt{2}(a, a, a)$ is 6 times that of a . $\square$

4.7 Negative results, with exact scope

- The global operator-norm route certifies analyticity only in a disk that shrinks with the box, on the full and on the physical space and for every contour (Proposition 4.6); a limit of one method.
- The Jacobian $\rightarrow$ Yang–Mills chain reproduces the free two-gluon threshold of a shrinking-coupling box (34_); no bearing on the mass gap.
- The Navier–Stokes $\rightarrow$ Yang–Mills map (21_ §2): the zero-quotient sequence exists only along a path whose coupling tends to 0; at fixed coupling nothing follows, and whether a different state with vanishing quotient exists is open.

- Theorem 4.1 and Corollary 4.5 hold only at strong bare coupling (with audited inputs from $g^2 \geq 4.0187$, conditionally from $g^2 \geq 3.6836$), and the standing continuum path leaves that domain.
- The constraint-data encoding is not a dynamical correspondence.

5 Where the workbenches meet

This section states each intersection found between the workbenches, and with the split-zero zeta programme, as a typed map: its domain and codomain, what crosses and what does not, and its status. The types are: P, a proved map or lemma used on both sides; I, the same identity or object on both sides; D, one side imports the other's theorem as an unproved dependency; R, use as a regression test only; A, an analogy or proposal; M, methodology only. Audit note: 47_.

5.1 The graph

The nodes are the zeta programme (zeta), Erdős–Straus (ES), Collatz (CZ), Erdős 817 (EP), Yang–Mills (YM), Navier–Stokes (NS), S^6 , and the Jacobian map F .

#	Edge	What crosses	Type	Status
1	zeta $\rightarrow$ CZ	the zeta programme's Hurwitz zero-cluster reconstruction (<code>historical_hurwitz_jets.tex</code> , H10–H13), re-proved in the part used in the Collatz note (§5.2.1)	P	audited
1a	zeta $\rightarrow$ CZ	the split-zero support modules and killed-representative quotient, instantiated on the Collatz operator (Collatz Ch. 7–9, 15)	P/D	located
2	NS $\rightarrow$ ES	the torus endomorphism $J = \begin{pmatrix} 3 & 1 \\ 1 & 5 \end{pmatrix}$ of the NS manuscript, as an operator only	P	audited (21_§1)
3	$S^6 \rightarrow$ ES	the monodromies A_1, A_2, A_∞ of the $(3, 4, \infty)$ period system (§1.5)	P	audited, odd levels D
4	$S^6 \rightarrow$ YM	the imaginary period block B of $\Pi(z)$, only through $\det B$ on the non-wrapping patch used by the workbench	P	audited (28_)
5	$F \rightarrow$ NS kinematics $\rightarrow$ YM	incompressible flows, the escaping curve, the material tensor, Theorem 14.2	P	audited; conditional on fixed-box limits
6	$F \rightarrow$ zeta	an inverse-fibre chart, fibre heat, a material generator, endpoint jets	P within zeta; A relative to YM	located
7	ES $\leftrightarrow$ zeta	a four-dimensional Keller map P with $\det DP = -2$ and its conductor; the shared file is byte-identical	P	$\det DP$ audited; rest located
8	zeta $\rightarrow$ YM	the Gram sandwich from a relative column error, $\delta \leq 1$	P	audited (21_§3)
9	zeta $\rightarrow$ YM	an inverse-power observability pattern	P	audited in part (its arithmetic)
10	zeta $\rightarrow$ YM	a composed minimum-lift identity and a Pythagorean section identity	P	checked numerically

#	Edge	What crosses	Type	Status
11	NS $\rightarrow$ YM	velocity $\mapsto$ reducible $SU(2)$ connection, $-2 \sum \text{tr } F_{ij}^2 = \lambda^2 \text{curl } u ^2$	P; D	audited; conditional on imported NS rate bounds and on YM sections not read
12	NS $\rightarrow$ zeta	the NS field as arithmetic input (Mellin transports)	D; P	located
13	$S^6, F \rightarrow$ NS (in zeta)	fluid coordinates; S^6 gauge-scalar equations $\rightarrow$ NS on $\mathbb{T}^3$	P	located
14	zeta $\rightarrow$ ES	the zeta results NG20–NG21 as theorem dependencies of ES item 20 and family C	D	located
15	ES $\rightarrow$ zeta	the 488-page ES reader vendored as a source dependency	D	located
16	EP $\rightarrow$ zeta	finite word sources in the supported quotient; a “cocycle” identity	P	one of ten interface notes audited (a restatement)
17	ES $\leftrightarrow$ CZ	$3(4n + 1) + 1 = 4(3n + 1)$	I	proved here (one line)
18	CZ $\leftrightarrow$ ES $\leftrightarrow$ zeta	the homogenisation $(x, h) \mapsto (ax + bh, h)$	I	audited in part (the ES diagram)
19	CZ $\rightarrow$ YM	$X_v - X_u = (q - 3)(q + 4)/32$	R	proved here (one line)
20	ES $\leftrightarrow$ zeta	the semiring $G(R) = R \sqcup \{\tau\}$	I	audited (21_§6)
21	ES $\leftrightarrow$ zeta	a finite Weyl basis of $\text{End } \mathbb{C}[A]$	I	audited in part (the ES side)
22	ES $\leftrightarrow$ S^6	the Niemeier lattice with root system $A_5^4 D_4$, glue index 72	I	audited in part (the ES code)
23–25	various	the word “gap”; a proposed success projector; validators and design documents	A, M	located
26	NS $\leftrightarrow$ zeta	the Mellin abscissa of the NS terminal singularity	A	a reading (21_)
26a	YM $\leftrightarrow$ zeta	the Yang–Mills workbench’s summary of the zeta routes	A	located
27	zeta $\leftrightarrow$ ES	a routing proposal	A	located
28	zeta $\rightarrow$ ES	the 13–16 September split-zero packages	P/D	located

No mathematical edge was found between Erdős 817 and Erdős–Straus, Navier–Stokes or S^6 ; the 817 workbench’s mentions of the others are methodological, and its README says it is not an Erdős–Straus artifact.

5.2 The substantive intersections

5.2.1 zeta $\rightarrow$ Collatz: the Hurwitz jets store moments (edge 1)

Fix the K odd-return words with residues r_w and a nontrivial zero ρ of ζ (§2.6). The map from probability vectors λ to the order- J jet of the zero cluster of $\sum_w \lambda_w \zeta(s, 1 + tr_w)$ near ρ factors as

$$\lambda \mapsto (M_1, \dots, M_J), \quad M_i = \sum_w \lambda_w r_w^i \mapsto (c_1, \dots, c_J),$$

a moment map depending only on the residues, followed by a triangular polynomial bijection whose coefficients are built from the derivatives at ρ of ζ and of the jet coefficients b_j (the diagonal coefficients being $\zeta'(\rho)$ and $b_j(\rho)$). The first factor is injective for $J = K - 1$ and not for $J = K - 2$ (Theorem 2.10). From zeta cross the reconstruction theorem and its ingredients (the jets

$\partial_t^j \zeta(s, 1+t)|_{t=0} = (-1)^j(s)_j \zeta(s+j)$, also the zeta reader's jet theorem, and $\zeta \neq 0$ on $\text{Re } s \geq 1$); from Collatz only the residues. Nothing about the location of ρ and nothing about orbits crosses. The zeta reader studies the same flow $t \mapsto \zeta(s, 1+t)$ at $t=0$ through its jets (the first jet vanishes in $-1 < \text{Re } s < 0$ exactly on the shifted zeros, and the flow does not translate the zero set); here the motion of one zero, taken at several shifts at once and averaged with weights, serves as storage. The Collatz workbench is the only one besides zeta found to use this flow.

5.2.2 NS $\rightarrow$ ES: an operator, not a fluid theorem (edge 2)

The auxiliary torus endomorphism J of the NS manuscript (determinant 14, eigenvalues $4 \pm \sqrt{2}$) enters 21 Erdős–Straus statements as an operator only. Their content is the identity $\{(u, v) \in G^2 : u^3 v = 1 = uv^5\} = \{(u, u^{-3}) : u^{14} = 1\}$ in a finite abelian group G (in the application the unit group modulo R_a), which gives exact aliasing on finite character groups, and the sharp constant $\inf_{k \neq 0} |v \cdot k| \|k\| = 1/\sqrt{4+2\sqrt{2}}$ for the unnormalised eigen-directions $v_r = (1, 1 - \sqrt{2})$, $v_t = (\sqrt{2} - 1, 1)$: the inequality is strict for every $k \neq 0$, and the infimum is approached along Pell vectors. No NS existence or blow-up theorem is used.

5.2.3 $S^6 \rightarrow$ ES and $S^6 \rightarrow$ YM: two uses of one period system (edges 3, 4)

Erdős–Straus uses the integral monodromies of the $(3, 4, \infty)$ period system to build affine-torsion covers of its divisor atlas (§1.5); Yang–Mills uses the imaginary period block B of the period matrix as a magnetic background, and on the non-wrapping patch used by the workbench only $\det B$ enters. Neither uses the global S^6 statement; both use displayed matrices.

5.2.4 The name “Fable”: one name, three objects (edges 5–7)

Fable is credited in Alpöge's announcement, as the workbench records it, for the work that produced the map F . In the repositories the name refers to three different objects: the three-dimensional Keller map F of the Jacobian-conjecture counterexample; the S^6 construction (in older files); and the four-dimensional Keller map P of the Erdős–Straus crosswalk. F enters Yang–Mills through incompressible flows and ends in the free two-gluon threshold (§4); it enters zeta through constructions written within zeta (located); P has $\det DP = -2$, and its fibres, monodromy and conductor are located.

5.2.5 zeta $\rightarrow$ YM: linear algebra, not arithmetic (edges 8–10)

The transfers state that no map between arithmetic zeta objects and gauge fields is asserted. What crosses is the Gram sandwich $(1-\delta)^2 \Phi^* \Phi \leq \Phi_J^* \Phi_J \leq (1+\delta)^2 \Phi^* \Phi$ when $\|(\Phi_J - \Phi)x\| \leq \delta \|\Phi x\|$ for all x and $\delta \leq 1$ (the zeta reader, §6; the zeta instance takes δ from a dilation factor, the Yang–Mills instance from e^{-Ta_0} for a heat semigroup); a left-inverse observability pattern; and standard Schur-complement identities. None carries an arithmetic constant into a Yang–Mills bound, and none enters Theorem 4.1.

5.2.6 NS $\rightarrow$ YM (edge 11)

A divergence-free velocity u becomes the reducible connection $A_i = \lambda u_i T$, with $-2 \sum_{i < j} \text{tr } F_{ij}^2 = \lambda^2 |\text{curl } u|^2$. Given the Yang–Mills inputs that the audit did not read (a free norm formula, an energy estimate and a convergence theorem), the resulting zero-quotient bound holds for every admissible curvature weight, of which the NS profile is one; the diagonal sequence exists only with coupling tending to 0, and at fixed coupling nothing follows (21_ §2).

5.2.7 Erdős 817 → zeta (edge 16)

Ten interface notes restate 817's finite constructions in the zeta programme's supported-quotient vocabulary. The one audited (the finite-period receiving map) is a correct, elementary restatement: free vector spaces on coefficient words, the evaluation quotient, the cut map $e_{(x,y)} \mapsto e_{x+Q(u)y}$ with $\dim \ker = F(u)F(v) - F(uv)$, and $\kappa(u,v)\kappa(uv,w) = \kappa(v,w)\kappa(u,vw)$ for $\kappa(u,v) = F(u)F(v)/F(uv)$, both sides being $F(u)F(v)F(w)/F(uvw)$. Nothing flows back into the 817 proofs, and nothing is claimed about ζ .

5.3 Shared identities

- *Erdős–Straus and Collatz* (edge 17). $3(4n+1)+1=4(3n+1)$, so n and $4n+1$ have the same odd successor. It is the workbench result R05 of Erdős–Straus (which credits the underlying diagrams to u/CivQ17) and the meeting-point step of the proof of Theorem 2.4. No recorded dependency either way.
- *Collatz, Erdős–Straus and zeta* (edge 18). The homogenisation $(x,1) \mapsto (x,h)$, $(x,h) \mapsto (ax+bh,h)$ makes affine branch maps commute with support inclusions. The zeta foundation cites it from the Clankers' Zenodo record 19900461; Erdős–Straus proves the diagram (X4); the Collatz note cites a local April 2026 note of the same title and uses its finite branch matrices.
- *Collatz and Yang–Mills* (edge 19). The constant terms $\frac{1}{2} + \frac{q}{32}$ and $\frac{1}{8} + \frac{q}{16} + \frac{q^2}{32}$ of two parity words differ by $(q-3)(q+4)/32$ and agree at $q=3$; Yang–Mills uses the identity as a regression fixture only.
- *Erdős–Straus and zeta* (edges 20, 21). In $G(R) = R \sqcup \{\tau\}$, $\{\tau\}$ is a prime ideal and $\{\tau, 0_R\}$ is prime iff R is a domain; the finite Weyl operators $T_a M_\chi$ form an orthogonal basis of $\text{End } \mathbb{C}[A]$.
- *Erdős–Straus and S^6* (edge 22). Both name the Niemeier lattice with root system $A_5^4 D_4$; $|\text{disc}(A_5^4 D_4)| = 6^4 \cdot 4 = 5184$, so the glue index is 72. The Erdős–Straus glue code (order 72, isotropic, cost spectrum $\{0:1, 4:46, 6:25\}$) was checked exhaustively; the S^6 side is only on Zenodo, so the two codes were not compared.

5.4 Three patterns that recur

Each pattern is a list of instances, each with the status it has above; nothing is claimed beyond the instances.

Exact restatements carry no new constraint. Theorem 1.1 and its re-encoding as sections of covers (§1.5); the Collatz conjecture as the vanishing of an abelian group, or of H_0 (§2); the Hurwitz encoding (Theorem 2.10); velocity data as gravitational constraint data (§4); 817's constructions in the supported-quotient vocabulary. Each is an equivalence or an injective map. An exact restatement moves a difficulty and does not reduce it unless the new side carries a constraint that the old side lacks; none of the instances exhibits such a constraint (for the Hurwitz encoding and the 817 interface there is no open problem being moved, only information being stored). The Collatz, 817 and Navier–Stokes sources say this of their own restatements; for the Erdős–Straus re-encoding the assessment is this audit's.

Method obstructions. A method or certificate family is shown unable to establish a property, usually by an explicit instance where the property holds and the method fails: the Erdős–Straus rows of §1 (every certificate prime is solvable); the global Neumann radius of Proposition 4.6, which shrinks with the box on the full and on the physical space and for every contour, whereas Theorem 4.1 and Corollary 4.5 obtain, on the physical space and for real ζ , a box-uniform gap by local norms, with audited inputs for $g^2 \geq 4.02$ and conditionally down to $g^2 \geq 3.68$; 817's modular certificates, which are sufficient but not necessary. The Collatz threshold, Theorem 2.2(c), is of a different kind: a limit on what the geometric model describes. The zeta reader's negative results have the same form as the first kind.

Regimes other than the problem's. Theorem 4.1 and Corollary 4.5 hold at fixed spacing and strong coupling; the low-mode theorem and the Navier–Stokes diagonal at coupling tending to 0 in growing boxes, where they reproduce free-field values; the regularity theorem for a modified Navier–Stokes equation; the Tao-clock laws, which are statements about first-entry laws in logarithmic density, along one scale sequence. Each workbench states its regime correctly.

What “certificate verified” means in each workbench:

Workbench	Certificate	What was re-done
ES item 22 817	polynomial positivity certificates modular progression-freeness of $H(B)$	recomputed with independent code recomputed twice
817 small values YM order 5	exhaustive searches 5,726 rational primal/dual certificates	four independent programs ($n \leq 6$) producer and auditor re-run, byte-identical; logic read
Collatz, Theorem 2.8	convergence of all odd $n \leq 330,749$	recomputed
Collatz, after Theorem 2.8	convergence of all $n < 2^{33}$	computed here (38 seconds)
Collatz, Corollary 2.9	convergence of all $n < 2.17 \cdot 10^{20}$ (Barina: 2^{68} by 2020, 2^{71} by 2025)	not re-run (a published computation)
Collatz, after Corollary 2.9	convergence of all $n \leq 3 \cdot 2^{69}$ (Barina, 2023), with Hercher's Corollary 29	not re-run; Hercher's proof not read
ES item 3	primality of a 22-digit prime	a Pocklington chain, computed here
ES item 4	1,381,117,764 rooted profiles	not re-run

5.5 Additions and corrections to the zeta reader's overlaps table

- The Collatz workbench has now been cloned (the zeta reader said it had not). The Hurwitz-jet encoding (edge 1) is a new, audited edge, and the split-zero support-module import (edge 1a) a new, located one.
- Edge 18: the zeta reader gave the Collatz side as the Clankers' Zenodo record; the Collatz note itself cites a local April 2026 note of the same title.
- $S^6 \rightarrow$ Erdős–Straus: the group theory and genus formulas are now audited, for odd levels.
- Erdős 817 $\rightarrow$ zeta: one of the ten interface notes is audited, as an elementary restatement.

Nothing here changes a statement of the zeta reader's Sections 2–7.

6 What was not checked, and points for the record

Erdős–Straus. The enumeration behind item 4; items 1, 5, 9–13, 15–20, 24, 25, 27, 28 beyond their statements; the deduction of item 23 from its certificates; the Pell and tail-sum sections of item 26; the packages ES-04 to ES-08 beyond one witness; X1; the ZIP and PDF archives (including the 488-page reader); the escape theorem of item 3.

Collatz. Chapters 4–8 and 10–19 beyond their theorem headings (except Theorems 8–9 of Chapter 6), and the proof of Chapter 9 itself; Chapter 2 §§4–6 and 8–10; Chapter 3 beyond §§1, 3–5 and its Theorem 6.1; Chapter 13's use of Matveev's bound; the large ledgers of Chapter 19 and its support budget; the predecessor-root claim of the $e = 8$ progression (only the workbench's checker was re-run); Tao's Proposition 1.11 and the preprint's repairs; the Siegel edition, the literature reader, the research companion (including a registered conjecture) and the Lean file (not built).

Erdős 817. The Lean builds and their 44 axiom reports; the workbench's own verifiers; the outer-control optimality (a 47,259-block audit); most of the cone-dual, recurrence, all-block-image

and seven-observable notes; nine of the ten interface notes; Costa's $k = 3$ preprint; the exact value of $g_4(7)$.

Yang–Mills, Navier–Stokes, S^6 . The logic of the fifth-order coefficient computation beyond re-running the workbench's programs; the order-3 and order-4 inputs (which Corollary 4.5(a),(b) do not use); several constants of the heat edition; the other editions YM-03 and YM-04, the proofs of YM-06 (whose gap theorems are located and compared with Corollary 4.5) and its cubic-residual input, the YM-05 write-up beyond its return section, and the research-control notes (YM-13) beyond their first lines; the imported Navier–Stokes theorem and NS-00 to NS-03, NS-05, NS-06; the existence step of NS-04(a), the Rindler response and the bootstrap of NS-04(d); S6-1 to S6-4, the literal part of S6-5, and the manuscript's statement that the threefold is diffeomorphic to S^6 .

Intersections. Edges 1a, 6, 12–15, 26a, 27 and 28 beyond locating them; nine of the ten 817 interface notes; the zeta side of edge 21 and the S^6 side of edge 22; the external record behind edge 18; the analogy and methodology edges beyond their statements.

Everywhere. Novelty: no literature search for novelty was made, except the reading of Korsky's preprint (Section 3) and a check of the Collatz verification and cycle bounds of Barina (the 2021 paper's publisher page; the 2025 paper's §§1, 2, 6 and Table 10; the project page), Hercher (the abstract and Corollary 29) and Eliahou (Section 2); Simons–de Weger and the physics literature cited in Section 4 were cited, not re-read.

Points for the record. These were found while surveying and are listed neutrally.

- The Yang–Mills files transcribe a zeta value, $W(wg_0, wg_0) = 2.0984855607004 \cdot 10^{-11}$, as “.10⁻¹⁹” in two places (ATTEMPTS.md line 274; research-attempts.json line 918).
- The S^6 attribution wording differs across files; CONTRIBUTION.md is the designated clarification.
- An Erdős–Straus handoff of 10 September says no Collatz repository was identified; the Yang–Mills research control of 14–15 September tracks the public Collatz workbench.
- The zeta programme's satellites 23, 26 and 27 credit Tao, Speyer and Poplett for the map F but not Alpöge's announcement, whose credit is supplied by the zeta README and a later file.
- The name “Fable” refers to three different objects (§5).
- Three proof files pinned in the Erdős–Straus README have changed since pinning, so the pinned line ranges drift by up to seven lines at the tip.

A How the audit was verified

First-pass audits. One Claude subagent per workbench (four in all) read the workbench read-only at the commits listed in the front matter, restated each result with its hypotheses, followed the proofs step by step and wrote independent checks, importing no workbench code; the Collatz and Yang–Mills first passes also re-ran the workbenches' own checkers, while the Erdős–Straus and 817 first passes wrote independent checks instead. Their reports are reproduced unchanged as 43a_–46a_ and an inventory of the Erdős–Straus, Yang–Mills, 817 and zeta repositories (made before the audits; the Collatz repository was cloned later, for its audit) as 47a_.

Re-runs. The first-pass Python scripts were re-run on 26 September by the author of this reader, except one 168-second script; the outputs are identical apart from timing lines. The first-pass C searches for g_4 were not re-run by the author; a separate search of the author's and, later, one of

the referee's replace them for $n \leq 6$, and the referee re-ran one of them. The Yang–Mills first pass had re-run the workbench's fifth-order producer (662 rows, 51 seconds) and its polynomial and dual-certificate auditor (20 seconds): both outputs are byte-identical to the published ones.

Independent checks. The script `checks/workbenches/workbench_reader_claims_checks.py` (47 items, all pass, about 7 seconds) tests the statements proved in this reader from their definitions alone: the shell count against brute force and the two sieves (Section 1); the ν_3 lemma, the merging families (600 pairs), the one-step cycles, the convergence of all odd $n \leq 330,749$, the least $m = 1636$ and $k \geq 679$, and the model inequality (Section 2); the digit set, the lift, κ_m , the certificates B^* and $A^\#$ and the composition identity (Section 3); α_5 , the threshold and d_5 by root isolation, the Casimir lemma on the cube, the face count, Proposition 4.7, the Kasner identities and the map F (Section 4). A third exhaustive search for g_4 (`g4_bitset.c`) found $g_4(6) = 79$ in 30 seconds. The generalisations of this reader are checked in `checks/workbenches/generality_checks.py` (items G1–G30, all pass, about 21 seconds): Theorem 1.1 for all odd primes below 260, for a up to $3p$, and its range $x < p/2$ on all solutions at primes below 400; Proposition 1.3 for every prime $p \equiv 1 \pmod{4}$ below $3 \cdot 10^4$, and Proposition 1.2 with its corrected form at $p \equiv 5 \pmod{12}$; Propositions 1.4 and 1.5; a Pocklington proof of the 22-digit prime of item 3; Proposition 2.1 and the bounds of Theorem 2.2(c) against the exact Δ ; Proposition 2.5 and its converse; the jet fibres of Theorem 2.10; the Eliahou-type bounds of Corollary 2.9 (which return Eliahou's 17,087,915 at 2^{40}), the bounds from the verification to 2^{33} (`descent_2e33.c`, 38 seconds) and the arithmetic of the published bound after it; Proposition 3.5 and the digit sets modulo 19; $\Lambda_3 = 3$ and the injectivity of ternary sums for 3-admissible sets; the M_n lower bound; Lemma 4.2 by exhaustive spin search and by linear programming; Corollary 4.5; the radii of Proposition 4.6; and Proposition 4.7 on $\mathbb{Z}^4$.

Referee passes. The notes 43_–47_ were refereed by a Claude instance that had written none of them (the twentieth pass of this audit). It re-derived every proof the notes present as complete and found no error in them; it reported 14 major findings (a hypothesis missing from the S^6 covers of Erdős–Straus, a false summary line, status labels stronger than the evidence, an incomplete credit, a false “only place” statement, an infimum stated as attained, and two intersections left out) and 56 minor ones. All were checked against the sources and applied before this reader was written.

This reader then had a referee pass of its own, by a further Claude instance that had written neither the notes nor the reader. It checked every written-out proof and found no error, recomputed the numbers with its own programs (among them a fourth exhaustive search for $g_4(n)$, $n \leq 6$, and a lex Gröbner basis showing that the fibre of F over $(-\frac{1}{4}, 0, 0)$ is exactly the three stated points), and spot-checked at least four statements per section against the repositories. It reported 8 major findings: an external claim about S^6 stated without attribution; one status label (NS-04(a)); three statements about how the audit was done; a sentence promising this paragraph; two scope conditions dropped in the intersections table (edges 4 and 11); one overstatement in §5's patterns; and an incomplete Erdős–Straus credit line. It also reported 52 minor ones (definitions, labels, locators, precision and typography, including the observation that Proposition 2.5 does not use one of the workbench's hypotheses). All were checked against the sources and applied.

At the owner's request the reader was then read for understatement and missed generality as well as for overstatement: first by its author, and then by a further Claude instance that had written none of it (the twenty-second pass). That pass reported 7 major findings: the gap theorem holds with audited inputs at truncation order 2 (Corollary 4.5); Corollary 2.9 needs only Barina's 2021 range; the least denominator of an Erdős–Straus solution is below $p/2$; the Casimir constant is the girth; the two-shell survivors and the reduction modulo 840 are theorems (Propositions 1.4 and 1.5); and the $k = 4$ construction lifts any admissible top level (Proposition 3.5). It also reported 11 minor ones, among them the converse in Proposition 2.5, the rates in Theorem 2.2(c), $\Lambda_3 = 3$, the physical-space radius of Proposition 4.6 and a primality certificate, and two overstatements:

the Neumann radii were compared across the full and the physical space, and YM-05 was said to depend on order 5. Each stronger statement was re-derived by the author of this reader and checked (G6–G30 above) before it was applied; the Barina 2021 range was checked against the 2025 paper and the project page. The results marked Generalised here come from these two readings.

A twenty-third pass, by a further Claude instance, then read the revised reader in both directions. It verified the new proofs and numbers with its own programs, and reported 5 major findings, each checked against the sources before being applied. One stronger statement taken over from the twenty-second pass was false: Proposition 1.2 does not hold as stated at $p \equiv 5 \pmod{12}$ (at $p = 5$, $E_a = \{2\}$ and $M_a = \{1, 4\}$), and the check claimed for it had tested only occupancy; the corrected counts are now stated and checked. Four were credit or scope: the published cycle bound of Hercher and Barina is about 1.9 times Corollary 2.9 and is now stated; Corollary 4.5(a) is the workbench's own YM-06 theorem, and YM-06 also has order-2 theorems; the rates in Theorem 2.2(c) and the uniformity in b are the workbench's own (4.3)–(4.4); and the M_n form of the linear lower bound is the workbench's (Theorem 3.7). It also reported 9 minor findings (among them Theorem 1.1(c) for every odd prime, part (a) of Theorem 2.2 for every $H \geq 0$, a proof of $\Lambda_3 = 3$ without citation, the per-member descent criterion, and two independent re-runs of the verification to 2^{33}), all applied.

Where everything is. The notes, reports, scripts and outputs are on the branch `claude/claude-ab-grind-20260925` of `KokunoYumeto/zeta-function-research-reader`, in `contrib/claude-ab/grind_20260925/` (notes `43_–47_`, `43a_–47a_`, and `checks/workbenches/`, which also holds the referee reports and their scripts).

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